

Part IV — The shop-visit market and the OEM aftermarket

Descriptive reference. Sources are cited inline and listed at the end. Takes no view.

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PART IV

The shop-visit market and the OEM aftermarket

This Part describes the market that Part I's engine creates: who takes CFM56 engines apart for a living, how many engines come through the doors each year, what stops the shops doing more, and how the engine's manufacturer earns money from every visit whoever performs it. A reader who finishes it will be able to read a capacity announcement, a shop-visit forecast and a parts-price statement with every unit and caveat understood, and will be able to lay out the four choices an operator has when an engine runs out of life.

TRANSLATION-TABLE ROWS ANSWERED

Part III §24 turns each fact about the fleet into a requirement on the aftermarket, one row per driver. This Part answers five of those rows. The row on low retirements (Part III §24, row T-1: low retirements) asks how many shop visits a year the retained fleet generates and for how long; §26 gives the forecasts, the base year and the retirement assumption that moves them. The row on the MRO (maintenance, repair and overhaul) capacity shortage (Part III §24, row T-8: MRO capacity shortfall and turnaround) asks what limits the shops' output and how long an engine waits; §25 names the shops and their announced additions, and §27 gives the slot, labour, parts, stand and turnaround figures. The row on OEM parts escalation (Part III §24, row T-7: OEM parts escalation) asks how the manufacturer's pricing reaches every visit; §28 gives the price steps and the service menu, and §29 shows which slice of a visit's bill they land on. The rows on LEAP (CFM's new-generation engine) durability (Part III §24, row T-3: LEAP durability shortfall) and on the GTF (Pratt & Whitney's geared turbofan) groundings (Part III §24, row T-2: GTF powder-metal groundings) create no CFM56 shop visits directly; they matter here because they keep old aircraft flying and because the new engines are sold under a different commercial model, which §28 sets beside the CFM56's. Part III is being written in parallel with this Part, so the row identifiers above are placeholders to be filled when its table is final. §30 closes with the four options an operator has for an engine at its life limit, which is where the rows meet.

25. Who performs shop visits: the OEM network, the independents, airline shops, FTAI

Part I §6 defined a shop visit as any removal of an engine for work in an engine shop, and an engine shop as a facility holding a Part 145 repair-station approval and, in practice, the manufacturer's data and tooling (Part I §9 has the approval). This section asks who owns those shops. The research sorted them into four groups: the manufacturer's own network, the independents, the airlines' in-house shops, and FTAI, which fits none of the first three neatly. The groups are described in turn, and the section ends with the one number the research most wanted and did not find, the manufacturer's share of CFM56 visits.

EXHIBIT 4.1 — WHO PERFORMS CFM56 SHOP VISITS: THE FOUR GROUPS, THEIR NAMED MEMBERS AND WHAT THE RESEARCH FOUND ABOUT EACH

Group	Named members reached by the research	What the group sells	What was found, and what was not
OEM network	GE Aerospace and Safran Aircraft Engines shops; licensed and joint-venture shops (often with an airline)	Shop visits under time-and-materials or flight-hour agreements; new parts to everyone	GE segment revenue and growth rates (\$28.7); no CFM56 shop-visit count for the network (unknown U3)
Large independents	StandardAero, MTU Maintenance, Lufthansa Technik, ST Engineering, AFI KLM E&M, Delta TechOps, Turkish Technic, GMF AeroAsia, EgyptAir Maintenance & Engineering, GA Telesis; also Aero Norway, FL Technics, Magnetic MRO, SR Technics	Shop visits priced to the customer, with material sourced from the OEM, used material, approved alternative parts and repairs	Capacity statements for StandardAero, MTU and FL Technics (Exhibit 4.2); none for the eight others (unknown U5)
Airline shops	The airline-subsidary MROs above when working on their parent's fleet; airlines maintaining their own engines	Visits for the parent fleet; some third-party work	No count of in-house CFM56 shops or of their share was retrieved
FTAI	Montréal (ex-Lockheed Martin Commercial Engine Solutions), Miami and Rome Fiumicino (QuickTurn), Lisbon, Orange; capacity partnerships with GMF AeroAsia and EgyptAir	Serviceable modules and engines by exchange or sale, rather than a visit priced to the customer (Part VII)	Capacity in shop visits (2022 language), module overhauls (2024, 2025) and modules (2026 call coverage); units differ (Exhibit 4.2)

Source: GE Aerospace, Form 10-K FY2025; Aviation Week, "Narrowbody Engine Demand Drives MRO Capacity Additions", 2025; StandardAero, Form 10-K FY2025; FTAI Aviation, Form 10-K FY2025 and releases of 9 Sep 2024 and 26 Feb 2025; compiled in D2 §3.1–3.4 and D5 §2.6. Reconciliation R-037, R-042; unknowns U-D2-03, U-D2-05.

25.1 THE OEM NETWORK

Part I §2 introduced CFM International as the joint venture of GE Aerospace and Safran Aircraft Engines, the manufacturer and type-certificate holder of the CFM56, and boxed the term "original equipment manufacturer", or OEM. Within the venture GE builds the core and Safran builds the fan, booster, low-pressure turbine and gearbox, and each partner sells spare parts for its own content. [D2 §3.1; CFM website, undated] Both partners also run shop networks of their own, and both place their names on the services menus described in §28.

GE reports the aftermarket inside one segment.

COMMERCIAL ENGINES & SERVICES (CES)

Commercial Engines & Services is the GE Aerospace segment that sells new commercial engines and spare parts and performs shop visits, in GE's own shops and under long-term service agreements. In 2025 it reported revenue of \$33,314 million, of which \$25,010 million (75.1%) was services, and segment profit of \$8,861 million, a 26.6% margin. The share of its services revenue earned on the CFM56 is not disclosed. Part XI §87 reuses these figures beside the other peers. [GE Aerospace, Form 10-K FY2025; D11 §3.8]

The segment's growth tells the reader how fast the shop-visit market has been moving at the manufacturer. GE's 10-K attributes 2025 CES revenue growth of \$6.4 billion (24%) to "increased spare parts volume, internal shop visit volume and worksopes, increased engine deliveries and pricing". [GE Aerospace, Form 10-K FY2025] Internal shop-visit revenue grew 24% in 2025, 19% in 2024 and 27% in 2023, and services revenue grew 26% in 2025. [GE Aerospace, Form 10-K FY2025] GE's first-quarter 2026 release headlined services revenue growth of 39%. [GE Aerospace, 1Q 2026 earnings release, Apr 2026] The 10-K also records a long-term materials agreement with Ryanair covering its entire fleet of about 2,000 CFM56 and LEAP engines (LEAP is CFM's new-generation engine, Part I §10), which §28.3 places on GE's services menu. [GE Aerospace, Form 10-K FY2025]

Safran reports the other half. Its 2025 group revenue was €31.3 billion, up 14.7%, with recurring operating income of €5.2 billion at a 16.6% margin. [Safran, 13 Feb 2026] Civil engine spare-parts revenue, which Safran tracks in US dollars, grew 17.6% in 2025, "driven primarily by the CFM56", on "a higher number of shop visits" and "a higher workscope content". [Safran, 13 Feb 2026; Forecast International, 13 Feb 2026] Safran had raised its civil aftermarket guidance twice in 2023 to growth in the "low-30% range", and raised full-year guidance again in October 2025 on what Leeham News called a "booming engine aftermarket". [Aviation Week, late 2023; Leeham News, 24 Oct 2025] Safran's installed base at end-2025 included "more than 22,800 CFM56s", from which it says it "can expect sustained aftermarket revenue from this fleet until 2030". [Safran, FY2025 results presentation, 13 Feb 2026]

The network is wider than the two partners' own buildings.

LICENSED SHOP; JOINT-VENTURE SHOP

A licensed shop is an engine shop that performs CFM56 work under a licence from the manufacturer; a joint-venture shop is one in which the manufacturer holds equity, often alongside an airline. Both operate with guaranteed access to the manufacturer's parts and technical data, and both are counted as part of the OEM network rather than as independents. The research did not retrieve a list of CFM56 licensed and joint-venture shops or their throughput. [D2 §3.1, §8]

GE's own description of the CFM56 market is that "an open network exists with multiple providers who compete and invest to win shop visits", and that GE will "continue to strengthen MRO access to OEM materials to support further CFM56 longevity". [GE Aerospace investor-relations page, 2026] Specific CFM56 shop-visit counts for the OEM network, and therefore the network's share of the market, were not found. [D2 U3] §25.6 returns to that gap.

25.2 THE LARGE INDEPENDENTS

Part I §6 boxed "MRO" as the industry's generic term for maintenance, repair and overhaul and "independent MRO" as an engine shop not owned by the engine's manufacturer. Three kinds of company fit that definition. Stand-alone companies such as StandardAero, Aero Norway and Magnetic MRO do nothing else. Airline subsidiaries such as Lufthansa Technik, AFI KLM E&M, Delta TechOps, Turkish Technik, GMF AeroAsia and EgyptAir Maintenance & Engineering maintain their parent's fleet and sell to third parties. And one engine maker's shop is independent on this engine: MTU Maintenance belongs to MTU Aero Engines, which is a partner in the V2500 (the engine that competes with the CFM56-5B on the A320ceo, Part I §10) but not in CFM, so on the CFM56 it is a customer of the manufacturer like any other shop. [D2 §3.2]

StandardAero is the independent the research knows most about, because it files with the SEC. Its Engine Services segment reported 2025 revenue of \$5,354.0 million, up 15.3%, and segment Adjusted EBITDA (earnings before interest, tax, depreciation and amortisation) of \$706.9 million, on StandardAero's own definition of the measure, a 13.2% margin; the segment covers commercial, business-aviation and military engines, and the filing does not give a CFM56 share. [StandardAero, Form 10-K FY2025; D11 §3.6] Its first-half 2026 report gave revenue of about \$3.2 billion, up 8.8%, and Adjusted EBITDA of \$433.0 million, up 12.3%. [StandardAero, Form 10-Q for the quarter ended 30 Jun 2026]

CENTER OF EXCELLENCE

A Center of Excellence is a shop or line dedicated to one engine type, so that tooling, training, test-cell correlation and parts flow are specialised to it. StandardAero's CFM56-7B Center of Excellence in Dallas, Texas, opened in the second half of 2024 "to service the growing demand on that platform". Part XI §87 uses the term when it profiles StandardAero. [StandardAero, Form 10-K FY2025]

StandardAero states that it has spent "more than \$100 million since 2022 to expand" and to "more than double our shop visit capacity on the CFM56-7B" through the Dallas Center of Excellence, and more than \$100 million on a LEAP line at its 810,000-square-foot San Antonio facility. [StandardAero, Form 10-K FY2025] Its Winnipeg expansion added 70,000 square feet, about 40% more footprint, and came online on 3 September 2025 with capacity for the CF34 (a regional-jet engine) and the CFM56. [GlobalAir, Sept 2025] The LEAP and CFM56 Dallas

programmes "reached profitability" in a 2025 quarter, according to the company's third-quarter 2025 filing. [StandardAero, Form 10-Q for the quarter ended 30 Sep 2025] Its chief commercial officer named labour as the leading constraint. [Aviation Week, "Narrowbody Engine Demand Drives MRO Capacity Additions", 2025]

MTU Maintenance overhauls the CFM56 and the V2500 at Hannover and at Zhuhai in China, and has a Berlin-Brandenburg shop; Aviation Week reports it "invested heavily in ramping up capacity, with expansions" at all three. [Aviation Week, 2025] MTU's commercial MRO business reported 2025 adjusted revenue of €6.0 billion, up 18%, and adjusted EBIT (earnings before interest and tax) of €478 million, an 8.0% margin, with geared-turbofan work about 40% of commercial maintenance; no CFM56 share was returned. [MTU Aero Engines, 24 Feb 2026; D11 §3.9] EME Aero, the MTU and Lufthansa Technik joint venture, is a geared-turbofan shop only and targets 500 interventions a year from 2028. [Aviation Week, 2025]

Lufthansa Technik runs a CFM56 engine shop at Hamburg. Its 2025 revenue was €8.049 billion, up 12%, and adjusted EBIT was €603 million, a 7.5% margin; Engine Services "generates almost half" of revenue. No CFM56 capacity or shop-visit figure was returned. [Lufthansa Technik, FY2025 results release, early 2026; D11 §3.9] ST Engineering, the Singapore-based group, named skills shortages and supply chain as its biggest challenges and gave no capacity figure. [Aviation Week, 2025] The three margins just quoted sit on three different bases, segment Adjusted EBITDA for StandardAero and adjusted EBIT for MTU and Lufthansa Technik, and Part XI §88 explains why they cannot be set side by side without saying so. [D11 §2.3]

The smaller specialists fill in the picture. Aero Norway, a CFM56 specialist at Stavanger, wrote in May 2024 that long lead times for induction slots were driving interest in quick-turn services, "especially in the case of older-generation engines, such as the CFM56 and V2500", and it has awarded a repair-management contract to GE (§28.5). [Aero Norway in AVM (a trade magazine), May 2024; GE Aerospace release, undated] FL Technics expanded its Kaunas engine shop for CFM56 demand and reports that about 70% of its work is on the -7B, the remainder on the -5B and the older -3. [FL Technics release, 2025] Magnetic MRO, based in Tallinn, discussed CFM56 "market challenges, opportunities" for 2024 in the trade press. [Aviation Week, 2024] SR Technics in Zurich publishes a CFM56-7B capabilities and price catalogue; a version dated 3 June 2026 was located and not read, which is why no shop's price list appears anywhere in this primer. [SR Technics, 3 Jun 2026; D2 §3.2]

For AFI KLM E&M, Delta TechOps, Turkish Technic, GMF AeroAsia, EgyptAir Maintenance & Engineering and GA Telesis the research established only that each operates a CFM56 shop or a related business. GMF AeroAsia and EgyptAir are FTAI's capacity partners (Part VII §50). EgyptAir holds a GE TrueChoice overhaul-consulting agreement (§28.5). GA Telesis is primarily

a trader in used material with an engine shop attached (Part V §35). No capacity or shop-visit figure was returned for any of the six. [D2 §3.2, U5; Amwal Al Ghad, undated]

25.3 AIRLINE SHOPS

The third group is the hardest to see. Many airlines maintain their own engines in their own shops and never sell a visit to anyone. Where an airline's maintenance arm has become a business in its own right it appears in the previous subsection, since Lufthansa Technik, AFI KLM E&M, Delta TechOps, Turkish Technic, GMF AeroAsia and EgyptAir Maintenance & Engineering are all airline subsidiaries. The in-house-only shops behind them were not profiled. The research retrieved no count of airline CFM56 engine shops, no estimate of the share of visits performed in-house, and no figure for how much of that work has moved to third parties as fleets have aged. [D2 §3.2, §3.4] The reader should treat the airline group as present in every forecast of §26 and absent from every capacity table of this Part.

25.4 FTAI

FTAI's business on the CFM56 is the Module Factory, which Part I §3 quoted from the FY2025 annual report and which Part VII §46 treats in full. It is listed here as a performer of shop work because its shops take engines and modules apart, repair them and test them, and because its capacity statements sit in the trade press beside the independents'. It differs from the other three groups in what it sells: a serviceable module or engine by exchange or sale, rather than a visit priced to a customer's own engine. [FTAI 10-K FY2025; D5 §2.2]

LMCES; MONTRÉAL

Lockheed Martin Commercial Engine Solutions (LMCES) was the Montréal engine shop that FTAI acquired from Lockheed Martin Canada, closing on 9 September 2024 for total consideration of \$170.0 million; the shop is about 500,000 square feet in the annual report and 526,000 in the closing release. It is the base of the Module Factory. Part VII §50 carries the facility in full. [FTAI release, 9 Sep 2024; FTAI 10-K FY2025; reconciliation R-038, R-039]

QUICKTURN; QUICKTURN EUROPE

QuickTurn is the brand of FTAI's Miami engine business, formed from the assets of iAero Thrust acquired on 4 January 2023 with Unical Aviation as partner; FTAI bought Unical's remaining interest on 1 December 2023 for \$30.3 million in cash. QuickTurn Europe is the 50% joint venture at Rome Fiumicino Airport, Italy, in the former IAG (International Airlines Group) Engine Center, announced 26 February 2025 and closed 5 June 2025 for \$10.5 million, with 200,000 square feet. Lower-case "quick-turn" is a workscope (Part I §6); the capitalised names are these businesses. Part VII §50 carries both. [FTAI releases, 1 Dec 2023, 26 Feb 2025 and 5 Jun 2025; FTAI 10-K FY2025; registry C-20]

FTAI's annual report lists 100%-owned maintenance sites at Montréal, Miami, Lisbon and Orange, the Rome joint venture, and a total of "over one million square feet"; its capacity partnerships with GMF AeroAsia in Indonesia and with EgyptAir were described on calls and their terms were not retrieved. [FTAI 10-K FY2025; D5 U8] Part VII §50 carries what is known about Lisbon and Orange; their deal terms, sizes and opening dates were not returned from any filing or release.

FTAI's capacity has been stated four times in four units, and each statement covers a different footprint. The annual report still carries language first used in 2022 describing Montréal as a facility "with a capacity for up to 300 shop visits per year". [FTAI 10-K FY2025; FTAI 10-K FY2022] At the Montréal closing FTAI said the Montréal and Miami shops together gave "capacity for 1,350 CFM56 module overhauls and >500 engine tests per year". [FTAI release, 9 Sep 2024] With the Rome agreement it said QuickTurn Europe "adds capacity for 450 modules (150 engines) per year, taking FTAI's total to 1,800 modules (600 engines)" and more than 600 engine tests a year. [FTAI release, 26 Feb 2025] On the second-quarter 2026 call management said it had increased CFM56 module production capacity to 3,000 a year, "supporting its 25% market share goal and 100 Mod-1 units annually"; that figure comes from call coverage and was not returned from the release text. [GuruFocus and Quartr coverage of the FTAI call of 30 Jul 2026; orchestrator notes] Lisbon's "113,000 square feet, 300+ modules a year" is also call coverage and unverified against any primary source; the annual report lists Lisbon as owned and the July 2026 call described it as "still ramping". [reconciliation R-037, R-041]

Two cautions govern how these figures are read. First, a shop visit, a module overhaul and a module are different units; FTAI's own releases convert at three modules per engine (450 modules for 150 engines, 1,800 for 600), which matches the three gas-path modules of Part I §3, and no Part of this primer compares the 300-visit figure with the module figures directly. [reconciliation R-035, B-3; registry C-4] Second, the figures are dated statements about growing footprints, not a single capacity revised over time. Exhibit 4.2 carries them beside the independents' statements with the unit each one counts.

Output is reported in modules. FTAI refurbished 757 CFM56 modules in 2025 against a target of 750, 184 in the second quarter of 2025, 228 in the fourth, 296 in the second quarter of 2026 (up 61% year on year) and 566 in the first half of 2026. [FTAI releases, 29 Jul 2025, 25 Feb 2026 and 29 Jul 2026; call coverage, Feb 2026] The 2026 target was set at 1,050 modules with the full-year 2025 results in February 2026 and was raised to 1,200 on the July 2026 call according to call coverage. [FTAI release, 25 Feb 2026; GuruFocus, Nasdaq.com and MarketBeat coverage, 30 Jul 2026; reconciliation R-036]

The research did one piece of arithmetic to size this against the market, and it carries three caveats. At three modules per engine, 1,200 modules is about 400 engine-equivalents, which is about 17% of a 2,350-visit year; 3,000 modules would be about 1,000 engine-equivalents, about

43%. [derived in the research, D2 §3.3] These are not shop-visit shares: modules are sold individually, many go into engines that also visit other shops, and the denominator FTAI uses for its "25% market share goal" was not retrieved. [D5 U5; reconciliation R-042] A third framing, from FTAI's own release of 26 February 2025, is a \$22 billion total addressable market for maintenance, repair and exchange in which FTAI held about 5% as of 2024. [FTAI release, 26 Feb 2025] Dollars, visits and module-equivalents are three different denominators, and Part VII §49 carries all three.

25.5 CAPACITY STATEMENTS AND ADDITIONS, 2022 TO 2026

Before the exhibit, the unit needs a definition, because the independents and FTAI count different things.

CAPACITY (SHOP)

A shop's capacity is the number of engine inductions a year it can take. Four things set it: floor space and engine stands, which bound how many engines can be open at once; licensed technicians; test-cell hours, since every rebuilt engine must be run before release; and parts flow, because a shop that cannot get turbine blades or repaired cases on time fills up with half-built engines that consume stands and floor without producing output. Additions are usually announced in square feet or as a multiple of previous throughput. FTAI's "modules per year" is a different unit, and Part VII §49 states the conversion FTAI's releases use. [D2 §2.4, §5; registry C-4]

EXHIBIT 4.2 — NAMED CAPACITY STATEMENTS AND ADDITIONS, 2022 TO 2026, WITH THE UNIT EACH COUNTS

Shop or site	Statement or addition	Unit counted	Date	Source and level
StandardAero, Dallas (CFM56-7B Center of Excellence)	">\$100 million since 2022"; "more than double our shop visit capacity on the CFM56-7B"; opened second half of 2024	Shop visits (as a multiple) and dollars	Feb 2026 (filing)	StandardAero 10-K FY2025 (filing)
StandardAero, San Antonio (LEAP line)	">\$100 million"; 810,000 sq ft site; LEAP-1A and -1B test-cell correlations completed 2024	Dollars and square feet	Feb 2026	StandardAero 10-K FY2025 (filing)
StandardAero, Winnipeg	+70,000 sq ft (about +40% footprint); CF34 and CFM56; online 3 Sept 2025	Square feet	Sept 2025	GlobalAir (trade press)

Shop or site	Statement or addition	Unit counted	Date	Source and level
MTU Maintenance, Hannover, Zhuhai, Berlin	"Invested heavily in ramping up capacity, with expansions"; CFM56 and V2500 at Zhuhai	None given	2025	Aviation Week (trade press)
EME Aero (MTU and Lufthansa Technik JV)	500 interventions a year from 2028; geared turbofan only	Interventions (visits)	2025	Aviation Week
FL Technics, Kaunas	Expanded for CFM56 demand; about 70% of work CFM56-7B	None given	2025	FL Technics release (company)
Safran, Querétaro, Mexico	New maintenance shop opened 30 June 2026	None given	Jun 2026	Safran release (company)
FTAI, Montréal	"Capacity for up to 300 shop visits per year"	Shop visits	2022 language, still in FY2025 10-K	FTAI 10-K (filing)
FTAI, Montréal plus Miami	"Capacity for 1,350 CFM56 module overhauls and >500 engine tests per year"	Module overhauls; engine tests	9 Sep 2024	FTAI release (company)
FTAI, adding Rome Fiumicino	" +450 modules (150 engines)"; total "1,800 modules (600 engines)"; ">600 engine tests"	Modules; engines at three per engine; tests	26 Feb 2025	FTAI release (company)
FTAI, all sites and partners	"3,000" CFM56 modules a year	Modules	Jul 2026	Call coverage (GuruFocus, Quartr); not in release text
FTAI, Lisbon	"113,000 sq ft, 300+ modules/yr"	Square feet; modules	2026	Call coverage; unverified
FTAI with GMF AeroAsia and EgyptAir	Capacity partnerships	None given	2026	Calls; terms not retrieved (D5 U8)

Shop or site	Statement or addition	Unit counted	Date	Source and level
Lufthansa Technik, AFI KLM E&M, Delta TechOps, Turkish Technic, ST Engineering, GA Telesis	No CFM56 capacity figure returned	None	n/a	D2 U5

Source: StandardAero, Form 10-K FY2025; GlobalAir, "StandardAero's Winnipeg expansion adds CF34 and CFM56 engine capacity", Sept 2025; Aviation Week, "Narrowbody Engine Demand Drives MRO Capacity Additions", 2025; FL Technics release, 2025; Safran release, 30 Jun 2026; FTAI Aviation, Form 10-K FY2025 and FY2022; FTAI releases of 9 Sep 2024 and 26 Feb 2025; GuruFocus and Quartr coverage of the 30 Jul 2026 call. Reconciliation R-037, R-041; E.1 rules 2 and 17; E.2 "Capacity".

Two patterns are visible in the table and both are about units rather than facts. The independents announce money and floor space and, once, a multiple of previous throughput; none of them states inductions a year. FTAI states modules, engines and tests, and once, in old language, shop visits. The only way to compare the two sets is to convert FTAI's modules to engines at the convention its own releases use, and even then an engine-equivalent of modules is not an induction. [reconciliation R-037; registry C-4] The table is therefore a list of dated announcements, not a capacity model, and the research built no such model. [D2 §4.8]

25.6 THE OEM SHARE: NOT KNOWN

No source the research reached gives the share of CFM56 shop visits performed in the manufacturer's network against the independents and the airlines. [D2 U3] Three facts bound the question without answering it. Safran describes the CFM56 aftermarket's revenue as 85% spare parts and time-and-materials work, so most visits are not performed under the manufacturer's flight-hour contracts (§28.2). [Safran via Aviation Week, c. 2023/24] Trade press puts about 15% of the CFM56 fleet under long-term agreements against about 50% of the LEAP fleet; the two figures are complements of the same view and are both Aviation Week's reporting of Safran. [Aviation Week, undated; reconciliation B-4] And Safran attributes part of its LEAP spares growth to "the increasing number of shop visits performed by third-party MROs", which implies the manufacturer tracks third-party share internally without disclosing it. [Safran, 13 Feb 2026; D2 §3.4] What would resolve the question is shop-visit-by-provider data from Aviation Week or Cirium, or a provider split in the Oliver Wyman fleet forecast. [D2 U3]

26. Demand: shop visits per year, the peak, and the retirement assumption behind it

Part III §23 carries the forecast curve as the manufacturers and forecasters publish it, and this section reads the same figures as a market: how many engines a year the shops of §25 can expect, when the count stops rising, and which single assumption moves it. The two sections cross-reference each other and neither re-boxes the other's terms. [registry, Part IV rows]

26.1 THE BASE YEAR

The reference point is 2019. Safran, as reported by Aviation Week, put annual CFM shop visits in 2019 at "just above 2,000", "almost all of them CFM56-5B and -7B". [Safran via Aviation Week, "Safran Sees Leap Aftermarket Emerging By Mid-Decade", c. 2023/24] Shop-visit counts for 2020 to 2023 were not found year by year. [D2 U2] For 2024 Safran expected total CFM shop visits "topping 2,000 again", a figure that includes some LEAP visits. [Safran via Aviation Week, c. 2023/24] Those are the only pre-2025 points the research holds, and the gap between 2019 and 2024 is carried as an unknown rather than filled.

26.2 THE TWO FORECASTS AND THE PEAK

Two views of the next few years come from the two halves of the same manufacturer at two dates, and the primer carries both without choosing. [reconciliation R-006, R-007; E.1 rule 15]

EXHIBIT 4.3 — CFM56 SHOP VISITS PER YEAR: EVERY FIGURE THE RESEARCH REACHED, WITH POPULATION, SOURCE AND DATE

Period	Figure	Population as the source states it	Source and level	Date
2019	"Just above 2,000"	Annual CFM shop visits, "almost all" CFM56-5B and -7B	Safran via Aviation Week (OEM via trade press)	c. 2023/24
2020 to 2023	Not found	n/a	D2 U2	n/a
2024	"Topping 2,000 again ... with some help from the Leap family"	All CFM, including LEAP	Safran via Aviation Week	c. 2023/24
2024 to 2025	"40 per cent increase in shop visits"	Headline; scope (all engines or CFM56) not captured	IBA (an appraiser) via Aviation Week	2024

Period	Figure	Population as the source states it	Source and level	Date
2025	CFM56 visits grew "mid-single-digit", with "a higher proportion of full work scope shop visits"	CFM56	Safran FY2025 release (OEM)	13 Feb 2026
2025	About 2,300 CFM56 visits within about 9,000 engine shop visits of all types (about 8,800 in 2024)	CFM56 within all engines	Aviation Week 2026 Fleet & MRO Forecast, via ePlaneAI (trade press forecast)	2026
2025 to 2026 (earlier view)	Peak "at about 2,500 per year"	"CFM56-variant shop visits"	Safran via Aviation Week	c. 2023/24
2026 and 2027	"Entered the year projecting about 2,300"; "expanded the range to 2,300–2,400"; "probably closer to 2,400"	CFM56	GE Aerospace via Aviation Week, "CFM56 Overhaul Demand Remains Strong" (OEM via trade press)	2026
Through 2028	"2,300–2,400 range through 2028"	CFM56	Same	2026
Apex	"Around the 2027-2028 time period, followed by a gradual fade"	CFM56	Same	2026
2027	Peak in first shop visits	CFM56 and V2500 first visits only	Visual Approach Analytics (independent research, headline)	2025/26
After 2027	737-700 passenger MRO demand "to decline sharply"	One sub-fleet	Acumen Aviation (appraiser note)	18 Dec 2025
2030	"More than 4,000 shop visits annually, roughly split between Leaps and previous-generation models"	All CFM; implies about 2,000 CFM56 as one dossier reads it	Safran via Aviation Week	c. 2023/24
2028	LEAP aftermarket revenue reaches parity with CFM56	Revenue, not visits	Safran Capital Markets Day 2024, via Quartr	2024

Source: Aviation Week, "Safran Sees Leap Aftermarket Emerging By Mid-Decade" and "Leap Aftermarket Poised For Growth" (dates not visible, c. 2023/24); Aviation Week, "CFM56 Overhaul Demand Remains Strong, GE Aerospace Says", 2026; Aviation Week, "IBA predicts 40 per cent increase in shop visits from 2024 to 2025", 2024; Safran, 13 Feb 2026; ePlaneAI reporting the Aviation Week 2026 Fleet & MRO Forecast; Visual Approach Analytics, 2025/26; Acumen Aviation, 18 Dec 2025; Quartr, Safran Capital Markets Day (CMD) 2024. Compiled in D2 §4.1 and D9 §4.5. Reconciliation R-006, R-007; D2 T6.

Read the two manufacturer views together. Safran's earlier framing, from around 2023 or 2024, has CFM56 visits peaking at about 2,500 a year in 2025 and 2026 and then falling, so that by 2030 the CFM56 would be about half of a 4,000-visit CFM total. [Safran via Aviation Week, c. 2023/24] GE's 2026 framing has a lower and later shape: about 2,300 visits in each of 2026 and 2027 at the start of the year, a range of 2,300 to 2,400 and "probably closer to 2,400" later in the year, that range held through 2028, and an apex "around the 2027-2028 time period, followed by a gradual fade". [GE Aerospace via Aviation Week, 2026] The later view is both lower at its top and later in reaching it. The two are from the two CFM partners at different dates, and the research did not retrieve a statement from either partner reconciling them. [D2 T1; reconciliation R-007]

Two other figures in the exhibit do not fit either curve and are carried as unreconciled. IBA headlined a 40% increase in shop visits from 2024 to 2025, without the scope of the count being visible in what the research captured; if that were CFM56 visits on a 2024 base above 2,000, the 2025 figure would be about 2,800 or more, which is above both manufacturer curves. [IBA via Aviation Week, 2024; D2 T6] Safran's own 2025 report says CFM56 visits grew only mid-single-digit, with a higher share of full worksopes. [Safran, 13 Feb 2026] The most likely explanation recorded in the research is that the two figures describe different populations, but that is a conjecture, and the writer's instruction is to carry both. [D2 T6; E.1 rule 15]

The third shape in the exhibit is a different measure. Visual Approach Analytics dates the peak in first shop visits of the CFM56 and V2500 to 2027. [Visual Approach Analytics, 2025/26] A first visit is one engine's first removal, and the share of the fleet that has not yet had one is treated in §26.4. The Aviation Week forecast gives the context: about 2,300 CFM56 visits inside about 9,000 engine shop visits of all types in 2025, so the CFM56 is roughly a quarter of all engine shop work by count. [Aviation Week 2026 Fleet & MRO Forecast, via ePlaneAI]

A PLATEAU IS NOT A PEAK

The earlier manufacturer view describes a peak: a top of about 2,500 visits in 2025 or 2026 and a fall after it. The later view describes a plateau: 2,300 to 2,400 visits for three years and a "gradual fade" after 2028. The difference matters to a shop deciding whether to add a stand or a test cell, because capacity added into a peak is idle sooner than capacity added into a plateau, and the research records that a shop or module factory adding capacity now "is adding it into a plateau" on the manufacturer's own forecast. What moved the shape between the two dates was not the fleet, which only shrinks, but the assumption about how fast it shrinks, which §26.3 takes up. Where the fade begins, and how steep it is, depends on that same assumption. [D2 §5.7; GE Aerospace via Aviation Week, 2026]

What population each count describes is not stated in the sources. None of them says whether a module exchange performed without a full induction counts as a shop visit, whether visits to the older -3, -5A and -5C variants are included, or whether a quick-turn counts the same as an overhaul. [D2 §4.1] The reader should hold the counts as the sources' own and not as a single series.

26.3 THE RETIREMENT ASSUMPTION BEHIND THE CURVE

Part I §2 boxed the retirement rate and gave GE's sequence; Part III §19 is the full treatment with every denominator. Here the point is mechanical. No new -5B or -7B is being built, so the installed base falls by exactly the number of engines retired or converted each year. [D9 §2.2] Every engine that is not retired keeps flying and keeps accumulating cycles, and eventually it needs the work this Part describes. So the retirement assumption is the one input that sets both the height and the length of the shop-visit curve. [D9 §2.2; D2 §5.7]

EXHIBIT 4.4 — THE RETIREMENT ASSUMPTION AND THE FLEET'S MATURITY: EACH FIGURE WITH ITS BASE, VENUE AND DATE

Item	Figure	Base	Source and level	Date and venue
GE planning assumption for 2026, first	3–4% of the fleet	CFM56 engines	GE CFO via Aviation Week (OEM via trade press)	"Last year", per the 2026 article
GE planning assumption, revised	2–3%	CFM56 engines	Same	Start of 2026
GE planning assumption, current	"Something between 1.5% to 2%"	CFM56 engines	Same	2026
2025 actual	About 1.5%, broadly the same as 2024	CFM56 installed base	GE Q4 2025 results via Leeham News	22 Jan 2026
2026 expected	About 2%, "below the 2–3% range previously anticipated"	CFM56 installed base	Same	22 Jan 2026
1Q 2026 actual	"Below 1%"	CFM56, GE's 2026 outlook assumes about 2%	GE Aerospace investor-relations page (OEM)	2026

Item	Figure	Base	Source and level	Date and venue
Aircraft retirements 2024	"Fewer than 200", similar to 2023, 70% below the 2020 peak	All aircraft, scope not visible	IBA via AviTrader (appraiser)	24 Sep 2024
Aircraft retirements 2024	587	Cirium Fleets Analyzer, scope not visible	Via Informa forecast page (data provider)	2025
Fleet maturity, 2019	57% of CFM56s had no first shop visit; 21% had one	CFM56 engines	GE Aviation & GECAS (GE's former leasing arm) Investor Day (OEM)	18 Jun 2019
Fleet maturity, 2024	About 70% had seen zero or one shop visit	CFM56 engines	Safran media (OEM)	2024
Fleet maturity, 2026	"Over half" with no visit; "another quarter" with one	CFM56 engines	Safe Fly Aviation (consultancy blog)	2026

Source: Aviation Week, "CFM56 Overhaul Demand Remains Strong, GE Aerospace Says", 2026; Leeham News, 22 Jan 2026; GE Aerospace, "Recent events: your questions answered", 2026; AviTrader (IBA data), 24 Sep 2024; Informa Markets commercial forecast page (Cirium data); GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran media, 2024; Safe Fly Aviation, "CFM56 Engine Market Report 2026". Reconciliation R-004, R-005, R-012; E.1 rule 13.

The sequence in the first three rows is one source disagreeing with itself over time, and the research records it as the mechanism behind the raised shop-visit range rather than as an error. [D2 T3; reconciliation R-004] The two 2026 readings differ by half a point, about 2% in Leeham's account of the January call against 1.5% to 2% in Aviation Week's quotation of the chief financial officer, and the primer carries both with their venues. [E.1 rule 13] The research describes the forecast error as retirements "over-forecast by 1.5–2.5 points in 2025–26", which is the gap between the first assumption and the realised rate. [D2 §5.7] The two aircraft-count rows cannot be reconciled from what was captured, because neither source's scope is visible; Part III §19 carries that tension. [reconciliation R-005]

WORKED EXAMPLE 4.1 — HOW THE RETIREMENT ASSUMPTION MOVES THE NUMBER OF ENGINES THAT STILL NEED SHOP VISITS

Inputs. An installed base of about 24,000 CFM56 engines (sourced: Aviation Business News, Sept 2025) or about 23,000 (sourced: CFM website, undated); the primer carries both and never averages them (E.1 rule 14). Retirement rates taken as the midpoints of GE's stated ranges, 3.5% for "3–4%" and 1.75% for "1.5–2%" (midpoints assumed for illustration; ranges sourced: GE CFO via Aviation

Week, 2026), the 2025 actual of about 1.5% (sourced: Leeham News, 22 Jan 2026) and the first-quarter 2026 rate of "below 1%" (sourced: GE investor-relations page, 2026).

at 3.5% on 24,000: 840 engines a year retired (on 23,000: 805)
 at 1.75% on 24,000: 420 a year (on 23,000: 403)
 at 1.5% on 24,000: 360 a year (on 23,000: 345)
 at under 1% on 24,000: fewer than 240 a year (on 23,000: fewer than 230)
 engines retained each year against the first plan: $840 - 420 = 420$ (at 1.75%); $840 - 360 = 480$ (at 1.5%)
 over three years, 2024 to 2026: roughly 1,260 to 1,440 engines still flying that the first plan had retired

So the fall in the assumed rate keeps something like 1,300 to 1,400 engines in service over three years that the earlier plan had counted out, on the 24,000 base; the 23,000 base gives about 5% less. Against that, the manufacturer's shop-visit range widened by about 100 visits, from "about 2,300" to "2,300–2,400", roughly 4% (derived). The research did not retrieve the model the manufacturer uses to convert retained engines into visits, and the two figures are not expected to match year by year, because a retained engine visits a shop only when its own clocks run out (Part I §5). One consistency check is available: 2,300 visits a year on 24,000 engines is 9.6% of the base, which would imply an average interval of about ten years between visits if every engine were on a regular cycle; CFM56 intervals are usually shorter than that, so either the 24,000 includes spare and stored engines or a large share of engines are flying out their remaining life with no further visit planned (Part I Exhibit 1.8 carries this as a tension). Limits of the example: the midpoints are assumed; the two base figures are contested and may include spares and stored engines; retirements and visits are linked through engine-by-engine clocks that this arithmetic does not model. [Aviation Business News, Sept 2025; CFM website, undated; GE CFO via Aviation Week, 2026; Leeham News, 22 Jan 2026; GE Aerospace investor-relations page, 2026; D9 §2.2 and T1; reconciliation R-001, R-004, R-008]

The same number sits on both sides of the aftermarket. A retained engine is a future shop visit, and it is also an engine that is not torn down, so its parts do not reach the used-material market of Part V and its remaining life does not reach the green-time market of Part II §16. [D9 §2.2; D2 §5.4] Part III §24's row on low retirements is built on that double effect, and §27.4 and §30 of this Part show where it lands in a shop.

26.4 HOW MATURE THE FLEET IS

The second input to the curve is how many engines have never been to a shop.

FLEET MATURITY

Fleet maturity is the distribution of an installed base by the number of shop visits each engine has had: none, one, or more. GE reported in June 2019 that 57% of CFM56 engines had not had a first shop visit and 21% had had one; Safran reported in 2024 that about 70% had seen zero or one. A first visit on the latest -5B and -7B variants comes near the life limit and so combines a performance restoration with life-limited-part replacement (Part I §6), which makes it a heavier visit than a mature-run restoration. [GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran media, 2024; D2 §4.2]

The three maturity readings in Exhibit 4.4 point the same way and come from sources of different weight. GE's 2019 figure and Safran's 2024 figure are manufacturer materials; the 2026 figure, "over half" with no visit and "another quarter" with one, about 75%, is from a consultancy blog and is labelled so. [reconciliation R-012; E.1 rule 3] The research did not resolve how 70% in 2024 and about 75% in 2026 can both hold for a fleet that is only getting older, and records the pair as directionally the same with the later figure from the weaker source. [D2 T7]

Two facts sit beside the maturity figures without the research joining them. Trade sources say first-run -5B and -7B engines of the latest variants fly "almost to LLP limit", so their first visit is a life-limit-driven one (Part I §6). [FlightGlobal, undated; reconciliation B-7] And Safran attributes part of its 2025 spares growth to "a higher proportion of full work scope shop visits". [Safran, 13 Feb 2026] Whether the second is a consequence of the first was not stated by either source, and the primer places them side by side and stops.

27. Capacity: slots, labor, parts, stands, turnaround

§25.5 defined a shop's capacity as the number of inductions a year it can take, set by floor and stands, technicians, test-cell hours and parts flow. Between 2023 and 2026 each of those four limits was reported as binding somewhere in the CFM56 shop network, and this section gives the figure and date for each. It closes with turnaround time, which Part I §6 boxed and which is the quantity an operator actually feels. Every figure below is from a filing, a consultancy survey or the trade press, and the exhibits say which. [D2 §2.4, §5]

27.1 THE FOUR LIMITS, AND THE UNIT THEY SHARE

A shop's output is engines released per year. Each limit caps that output in a different way. Floor space and engine stands cap how many engines can be open at once; a shop that is full of engines waiting for parts is full, whatever its technicians could do. Technicians cap how many engine-hours of work can be done in a month, and the work is licensed, so a vacancy cannot be filled quickly. Test-cell hours cap how many rebuilt engines can be proven and released. Parts flow caps all three, because an engine waiting for a turbine blade or a repaired case occupies a stand,

has consumed labour, and cannot go to test. [D2 §2.4] The industry expresses an addition to any of these as square feet, as §25.5's exhibit showed, because the building is the limit that can be seen and announced. [D2 §5.5]

27.2 INDUCTION SLOTS

INDUCTION; INDUCTION SLOT

Induction is the acceptance of an engine into work at a shop, the point from which turnaround time is counted. An induction slot is the date a shop commits to start a given engine. When shops are full, the wait for a slot is distinct from, and additional to, the turnaround, and in 2023 to 2026 that wait became the binding constraint on an operator's engine availability. In other Parts "slot" also means a production slot for alternative parts (Part VI) or a reserved manufacturing slot for turbines (Part IX); this primer says "induction slot" for the shop meaning. [D2 §2.1, §2.4, §8; registry C-18]

Bain reported in 2024 that "the wait times just to secure an MRO slot rose by two to three months and, in some cases, six months", and projected that engine maintenance demand would exceed available capacity by more than 17% before the end of the decade if capacity kept its current trajectory. [Bain & Company, 2024] Trade press reported shops "at full capacity over the past year" in 2025. [Aviation Week, 2025] Aero Norway's account from the shop floor is that long lead times for induction slots were pushing operators toward quick-turn work, "especially in the case of older-generation engines, such as the CFM56 and V2500". [Aero Norway in AVM, May 2024] An engine-stand utilisation report carried by the trade press expects engine maintenance demand to peak in the near term in 2026 and to remain constrained through the end of the decade. [AviationPros, undated]

WHY THE SLOT, NOT THE TURNAROUND, BECAME THE BINDING CONSTRAINT

An operator planning a removal faces two waits in series. First it waits for a shop to accept the engine; then it waits for the shop to finish. The second wait is the turnaround time of §27.6, which lengthened from about 60 days to 90 to 120 days for a full overhaul. The first wait, which was close to zero when shops had spare stands, grew to two to six months. An aircraft without an engine is grounded or flies on a leased spare for both waits together, so the slot wait added more calendar time to a removal than the longer turnaround did. That is what raised the value of anything that avoids a full induction: a quick-turn visit, a module exchange from inventory, or a green-time engine flown in place of the one that ran out. Each of those substitutes shifts where and how the work is done; none of them changes how many engines reach their limits. [D2 §5.1; Bain, 2024; D9 §5]

27.3 LABOUR

Oliver Wyman's 2026 maintenance survey, of more than 150 respondents, found that "two-thirds of MRO operators struggle to find qualified technicians", that labour-rate inflation had "settled in

at 5.5–6.0%", and that "more than half of survey respondents expect labor attrition to be worse in the coming year". [Oliver Wyman, Apr 2026; ePlaneAI, 2026] StandardAero's chief commercial officer and ST Engineering both named labour as the leading constraint on adding capacity. [Aviation Week, 2025] Oliver Wyman also cites the retirement of skilled workers as a driver of rising maintenance cost, and no headcount or shortfall figure for engine technicians surfaced in the research. [Oliver Wyman via AviationPros, 2026; D9 U8]

Labour is 20% to 30% of a shop visit on one share breakdown and 15% to 20% on another (\$29.1), so a 5.5% to 6.0% rise in labour rates adds roughly 1.1 to 1.8 percentage points to the cost of a visit each year on its own. [derived in the research, D2 §5.2; Air Cargo Week, 2025/26; Safe Fly Aviation, 2026] Worked example 4.2 puts that beside the parts line. One labour data point from FTAI belongs here for scale: the company reported 985 full-time employees and independent contractors at 31 December 2025, of whom 494 were full-time employees in Canada, about 71% of them covered by collective bargaining agreements; the figure mixes employees and contractors, as the filing's wording does. [FTAI 10-K FY2025; E.1 rule 24] Part X §85 carries it.

27.4 PARTS

The parts constraint is the one the sources disagree about, and the disagreement is carried whole.

On one side is the manufacturer's own filing. GE's 10-K for 2025 reports that its operating system raised material input from priority suppliers by more than 40% year on year and that turnaround times for the LEAP, the CFM56 and the GE90 (a widebody engine) improved by more than 10% year on year in the fourth quarter of 2025. [GE Aerospace, Form 10-K FY2025]

FLIGHT DECK (GE'S OPERATING SYSTEM)

FLIGHT DECK is the name GE Aerospace gives its proprietary operating system for running its factories and shops. Its FY2025 annual report credits it with increasing material input from priority suppliers by more than 40% year on year, enabling a 26% rise in services revenue, and with the turnaround improvement quoted above. The research holds no description of what the system consists of beyond the name and those two claims. [GE Aerospace, Form 10-K FY2025; D2 §2.4, §3.1]

On the other side are the surveys and the trade press. Oliver Wyman's April 2026 survey found labour and material shortages remained the primary disruptors; two-thirds of respondents reported that next-generation narrowbody shop-visit costs were running 21% or more above expectation, and a quarter reported 50% or more. [Oliver Wyman, Apr 2026] Aviation Week reported in 2025 that "parts supply remains a capacity constraint with every OEM, due to the recovery in aircraft utilization and the associated growth in MRO demand coming at the same time that engine OEMs are ramping up production". [Aviation Week, 2025] AVM reported repair lead times "surpassing more than 200 days in severe cases" on some new-generation programmes, and the

trade press describes four-to-six-month waits for repairs at third-party facilities, with buyers advised to order life-limited parts and critical rotables early and to use exchange pools and used material. [AVM Magazine, 2025; Air Cargo Week and Locatory, Mar 2026] The same trade press says "the supply of used serviceable material remains thin, leaving shops more reliant on newly manufactured core parts", and Aviation Week ran a column headed "A Seller's Market for Used Parts". [AVM Magazine, 2025; Aviation Week, undated]

The two sides are not necessarily about the same thing. The manufacturer's statement is about its own suppliers and its own shops in one quarter; the survey is about operators' experience across all shops and programmes, and the cost-overrun figure is about new-generation engines. But the research records it as a tension because a reader will meet both: the manufacturer reports easing in late 2025, and the operator survey of April 2026 does not. [D2 T8; reconciliation R-020] What would narrow it is a dated lead-time series for turbine blades, castings and forgings, which the research sought and did not find. [D2 U6]

The used-material side of the parts constraint has a cause this Part has already named. Retirements of 1.5% to 2% rather than 3% to 4% mean fewer teardowns, fewer used parts and fewer green-time engines, so shops are pushed back to new parts at the escalated list price, and the manufacturer's pricing power rises. [D2 §2.4, §5.4] Part V §33 carries the supply side of that and Part VII §52 the agreement FTAI signed with CFM for parts (§28.8).

27.5 ENGINE STANDS AND FLOOR

ENGINE STAND

An engine stand is the wheeled fixture that holds a roughly 2.5-tonne engine, or a module, while it is worked on, stored or transported. Every engine in a shop occupies a stand from induction to release, so the stock of stands is a physical limit on work in progress, and a shortage of stands shows up as engines that cannot be inducted even when technicians are free. Utilisation of stands for legacy engines rose by about a third in 2025 over 2024, with CFM56-5A/B stand usage up 60%, according to one trade-press report. Part III §23 glosses the term at first mention. [D2 §2.1, §8; AviationPros, undated]

The stand figures are the only hard numbers the research holds on this limit. A report carried by AviationPros says engine-stand utilisation for legacy engines rose by a third in 2025 compared with 2024, and that CFM56-5A/B stand usage rose 60%. [AviationPros, "Engine Stand Utilization Report", undated] No figure for a stand shortage, a stand lead time or a stand price was found. [D2 U6] Floor space is reported as the proxy: StandardAero's Winnipeg addition of 70,000 square feet and FTAI's Lisbon site (per call coverage, unverified) are announced in square feet because stands and floor are the limit that an announcement can count. [D2 §5.5; GlobalAir, Sept 2025]

EXHIBIT 4.5 — CAPACITY CONSTRAINTS IN THE ENGINE SHOPS: THE FIGURES, WITH SOURCE, LEVEL AND DATE, AND BOTH SIDES OF THE PARTS QUESTION

Constraint	Figure	Source and level	Date
Slot wait	Up by two to three months, in some cases six	Bain & Company (consultancy)	2024
Demand against capacity	Demand to exceed capacity by more than 17% before 2030, if capacity grows on its current trajectory	Bain	2024
Shops' loading	"At full capacity over the past year"	Aviation Week (trade press)	2025
Technicians	Two-thirds of MRO operators "struggle to find qualified technicians"	Oliver Wyman survey, >150 respondents (consultancy)	Apr 2026
Labour-rate inflation	"Settled in at 5.5–6.0%"	Oliver Wyman	Apr 2026
Attrition	More than half expect labour attrition to be worse in the coming year	Oliver Wyman	Apr 2026
Next-generation shop costs	Two-thirds report 21% or more over expectation; a quarter report 50% or more	Oliver Wyman	Apr 2026
Parts, all OEMs	"Parts supply remains a capacity constraint with every OEM"	Aviation Week	2025
Parts, GE's own supply	Priority-supplier material input up more than 40% year on year	GE Aerospace 10-K FY2025 (OEM filing)	Feb 2026
Turnaround, GE's shops	LEAP, CFM56 and GE90 turnaround improved more than 10% year on year in 4Q 2025	GE Aerospace 10-K FY2025	Feb 2026
Repair lead times, severe cases	More than 200 days, some new-generation programmes	AVM Magazine (trade press)	2025
Third-party repair waits	Four to six months	Air Cargo Week and Locatory (trade press)	Mar 2026
Used material	Supply "remains thin"; "a seller's market for used parts"	AVM Magazine; Aviation Week	2025; undated
Engine stands	Legacy-engine stand utilisation up about a third in 2025; CFM56-5A/B up 60%	AviationPros stand report (trade press)	Undated

Constraint	Figure	Source and level	Date
Engine-stand shortage; blade, casting and forging lead times	Not found as numbers	D2 U6	n/a
Global MRO market	More than \$136 billion (2025); about \$140 billion (2026)	Oliver Wyman	Apr 2026
Engine MRO market	\$70 billion (2026) rising to \$102.6 billion (2036)	Oliver Wyman via AviationPros and Air Cargo Week	2026

Source: Bain & Company, "Get a step ahead of the engine maintenance capacity crunch", 2024; Aviation Week, "Narrowbody Engine Demand Drives MRO Capacity Additions", 2025; Oliver Wyman, "The new MRO supply paradigm", Apr 2026; GE Aerospace, Form 10-K FY2025; AVM Magazine, "The Engines Capacity Crunch" and "Inside the Engine MRO Supply Chain", 2025; Air Cargo Week / Locatory, Mar 2026; Aviation Week, "MRO Memo: A Seller's Market For Used Parts"; AviationPros, "Engine Stand Utilization Report". Compiled in D2 §4.7 and D9 §4.5, §6. Reconciliation R-020.

27.6 TURNAROUND, AND HOW IT DIFFERS FROM THE WAIT

Part I §6 boxed turnaround time (TAT) as the calendar time from induction to release and gave the headline figures. This subsection presents every turnaround figure the research reached, each with its source, because the figures come from different shops, workscopes and dates and the primer does not pick among them. [reconciliation R-019; E.2 "Turnaround"]

EXHIBIT 4.6 — TURNAROUND TIME FOR CFM56 WORK: EVERY FIGURE, WITH WORKSCOPE, SOURCE, LEVEL AND DATE

Workscope or scope	Turnaround	Source and level	Date
Full CFM56 overhaul, pre-pandemic	About 60 days	Aviation Business News / AVM (trade press)	2025
Full CFM56 overhaul, current	90 to 120 days	Same	2025
Light work	About 45 days	Same	2025
CFM56, GE Aerospace's shops	"Sustaining ~90 days"	GE Aerospace, Bernstein conference presentation (OEM)	27 May 2026
LEAP, CFM56 and GE90, GE's shops	Improved by more than 10% year on year in 4Q 2025	GE Aerospace 10-K FY2025 (OEM filing)	Feb 2026
Performance restoration	75 to 90 days	Safe Fly Aviation (consultancy blog)	2026
Heavy visit with LLPs	120 to 150 days	Safe Fly Aviation	2026

Workscope or scope	Turnaround	Source and level	Date
Legacy-engine turnaround, change over five years	Up 35%	Air Cargo Week (trade press)	2026
New-generation-engine turnaround, change over five years	Up more than 150%	Air Cargo Week	2026
Wait for an induction slot, additional to turnaround	Up by two to three months, up to six	Bain (consultancy)	2024
Module exchange from inventory	A quick-turn; FTAI's own exchange turnaround not retrieved	StandardAero menu (Part I §6); D5 U3	2022; n/a

Source: Aviation Business News / AVM, "Inside the engine MRO supply chain", 2025; GE Aerospace, Bernstein Strategic Decisions Conference presentation, 27 May 2026; GE Aerospace, Form 10-K FY2025; Safe Fly Aviation, "Engine Shop Visit Costs Worldwide 2026" and "CFM56-7B engine availability report 2026" (consultancy blog); Air Cargo Week, "From PW1100G to CFM56: the engine maintenance trends shaping 2026"; Bain & Company, 2024; StandardAero CFM56-7B brochure, Oct 2022. Reconciliation R-019; D5 U3.

The trade press attributes the lengthening since the pandemic to "long turnaround times for material repairs ... primarily due to parts scarcity issues in the MRO supply chain". [Aviation Business News, 2025] Air Cargo Week's figures say legacy-engine turnaround is up 35% against five years earlier and new-generation turnaround up more than 150%, which places the CFM56's problem as the smaller of the two. [Air Cargo Week, 2026] GE's own figures run the other way for its shops: about 90 days sustained for the CFM56 in May 2026 and an improvement of more than 10% in the fourth quarter of 2025. [GE Aerospace, 27 May 2026; GE Aerospace, Form 10-K FY2025] The consultancy figures sit between: 75 to 90 days for a restoration and 120 to 150 days for a heavy visit with life-limited parts. [Safe Fly Aviation, 2026] The figures are not reconciled because they describe different shops and worksopes, and the reader should carry the range.

The economic weight of turnaround is the spare engine. Every day an engine is away is a day the aircraft needs another engine or does not fly, and Part I §6 worked the rent for one restoration at two spare-engine rates, giving a range from about \$126,000 to about \$700,000 depending on the rate, the turnaround and whether a slot wait is included. [Part I, worked example 1.5; IBA, Apr 2024; Safe Fly Aviation, 2026] A module exchange from inventory aims to turn a 90-to-120-day event into a quick-turn of about 45 days, which is the hook Part VII §46 hangs FTAI's product on. [D1 §2.6; D2 §2.1] FTAI's own exchange turnaround was not retrieved. [D5 U3]

WORKED EXAMPLE 4.2 — WHAT A YEAR OF LABOUR AND PARTS INFLATION DOES TO THE PRICE OF ONE SHOP VISIT

Inputs. A performance-restoration visit at the midpoint of the \$1.2–1.6 million range, \$1.4 million (sourced: Safe Fly Aviation, consultancy blog, "2026 industry data"; this Part's range under E.1 rule 4), with the alternative midpoint of \$2.15 million from the same publisher's \$1.8–2.5 million range (sourced: Safe Fly Aviation, other page; Part I's range). Cost shares of material 60–70% and labour 20–30% (sourced: Air Cargo Week, 2025/26), with the alternative labour share of 15–20% (sourced: Safe Fly Aviation). Labour-rate inflation of 5.5–6.0% (sourced: Oliver Wyman, Apr 2026). Parts escalation of about 10% a year, the rate reported for new spares in 2022–23 (sourced: Aviation Week, late 2023), with the alternative of about 7% a year, the compounded rate on full life-limited-part sets 2008–2025 (derived in Part I, worked example 1.3). The 2024, 2025 and 2026 catalogue steps are not known (D2 U1), so both parts rates are illustrative.

labour line: $20\text{--}30\% \times 5.5\text{--}6.0\% = 1.1 \text{ to } 1.8$ points of visit cost a year
 (at the 15–20% share: 0.8 to 1.2 points)
 material line at ~10%: $60\text{--}70\% \times 10\% = 6.0 \text{ to } 7.0$ points a year
 material line at ~7%: $60\text{--}70\% \times 7\% = 4.2 \text{ to } 4.9$ points a year
 total, parts at 10%: 7.1 to 8.8 points a year
 total, parts at 7%: 5.3 to 6.7 points a year
 on a \$1.4M visit: \$74,000 to \$123,000 more each year
 on a \$2.15M visit: \$114,000 to \$189,000 more each year

So on these inputs a visit's price rises by five to nine percent a year before the shop changes anything about how it works, and about four-fifths of that rise comes from the material line that is priced by the manufacturer, not by the shop. That is the arithmetic behind the statement, carried through this primer, that catalogue escalation reaches every shop visit whoever performs it (\$28.1, \$29.5). Limits of the example: the shares are secondary figures that disagree with each other (reconciliation R-014); the parts escalation for 2024–26 is unknown and the two rates shown are from different years and part scopes (R-018); the repairs line (10–20%) is left at zero inflation for want of a figure; and the visit price ranges are themselves contested (R-013). [Safe Fly Aviation, 2026; Air Cargo Week, 2025/26; Oliver Wyman, Apr 2026; Aviation Week, late 2023; Part I worked example 1.3; D2 §5.2, §5.6]

28. The OEM aftermarket model: list prices, escalation, TrueChoice, EngineWise, long-term agreements

The manufacturer earns from a shop visit in two ways. It sells the new parts the visit consumes, at a published price it revises each year, to any shop that performs the work. And it sells services under its own brand: visits in its own shops, parts and repairs to other shops, and multi-year contracts under which it takes over the maintenance of an engine for a price per flight hour. The balance between those two is what distinguishes the CFM56's aftermarket from the LEAP's, and this section takes them in order.

OEM AFTERMARKET MODEL

The OEM aftermarket model is the way the engine's manufacturer earns from its installed base after delivery: spare parts sold at an escalating catalogue list price to every shop, plus branded services (shop visits, parts and repair packages, used material, repair management) and long-term agreements priced per flight hour. On the CFM56 about 85% of aftermarket revenue is spare parts and time-and-materials work, against about 40% for the LEAP, so the CFM56 is an open, transactional market in which the manufacturer's main capture is parts rather than contracts, and the LEAP is being set up as a largely contracted one. [Safran via Aviation Week, c. 2023/24; D2 §2.3, §3.4]

28.1 SPARE PARTS AT LIST PRICE, ESCALATING

Part I §4 boxed the catalogue list price as the manufacturer's published price for every new spare part, repriced at least annually, and escalation as the annual increase in it. The mechanics are simple and the consequence is large. The shop buys the part at list and passes it through at list plus a handling margin, or the airline buys the part directly under a materials agreement; either way the manufacturer receives the list price for every new part a visit consumes. [D2 §2.2] Used and alternative parts are priced as discounts to the same list, so they inherit its escalation (Part V §34, Part VI §44). [D3 §2.5]

EXHIBIT 4.7 — SPARE-PARTS ESCALATION STATEMENTS, BY DATE, SCOPE AND SOURCE

Date or period	Statement	Scope	Source and level
2008 to 2025	Full -7B life-limited-part set compounded at about 7% a year (\$1.775M in 2008; \$3.4M in 2018; \$5.7M in June 2025)	Life-limited parts only	Derived in Part I from Aircraft Commerce Issues 58 and 120 and MyAirTrade (trade press and market-data site)
Year not shown	One escalation step of about 12% on -5B/-7B life-limited parts; escalation set by "a formula which factors in labour costs, the price of raw materials and energy costs"	Life-limited parts	Ishka (trade publication), undated
1 Nov 2022	"Double-digit" increase	New spare parts generally	Aviation Week, late 2023 (trade press)
Aug 2023	"High-single-digit" increase by GE and Safran	New spare parts generally	Aviation Week, late 2023
2023	-7B parts up 20–30% after CFM's catalogue revision	-7B parts generally	Magnetic Group via Aviation Week, 2024

Date or period	Statement	Scope	Source and level
Trend to late 2023	New spare-parts prices rising "around 10% per year"	New spare parts generally	Aviation Week, late 2023
Stated rule	"Always been above inflation by 3 to 4 points ... we'll continue on that path, I would say, at least for some years"	Safran spare-parts pricing	Safran chief executive via Aviation Week, late 2023
Quarter reported in late 2023	GE external spares sales up 35% year on year; record \$42.2 million of spares shipped per day	GE spares	Aviation Week, late 2023
2024, 2025, 2026	Catalogue revision percentages not found	n/a	D2 U1; D1 U10 (cluster K-03)

Source: Aviation Week, "Parts Price Hikes Help Boost Safran, GE Aftermarket Sales", late 2023; Aviation Week, "Magnetic expects CFM56 market challenges, opportunities in 2024"; Ishka, "Pratt & Whitney mulls further LLP escalation hike", undated; Part I, Exhibit 1.6 and worked example 1.3. Reconciliation R-018; E.2 "OEM price escalation".

Two series run through the exhibit and they are not the same thing. One is specific to life-limited parts: a set price that compounded at about 7% a year over seventeen years, a single step of about 12% reported without its year, and a 20% to 30% rise in -7B parts after the August 2023 revision. [Part I worked example 1.3; Ishka, undated; Magnetic Group via Aviation Week, 2024] The other covers all spares: about 10% a year to late 2023, a double-digit step on 1 November 2022, a high-single-digit step in August 2023, and a stated policy of inflation plus three to four points. [Aviation Week, late 2023] They cover different parts, years and bases, and the primer reports both without combining them. [reconciliation R-018]

The gap in the record is recent. No percentage was found for the 2024, 2025 or 2026 catalogue revisions, so the only dated steps the primer can quote are the two of 2022 and 2023, and the only forward statement is Safran's rule. [D2 U1; cluster K-03] Coverage of each year's CFM price revision in Aviation Week or Aircraft Commerce, or a question on price in a GE or Safran earnings call, would close the gap. [D2 U1] In the same late-2023 reporting, GE's external spares sales were up 35% year on year and its spares shipment rate reached a record \$42.2 million a day. [Aviation Week, late 2023]

WORKED EXAMPLE 4.3 — COMPOUNDING THE TWO ESCALATION READINGS ON THE MATERIAL LINE OF ONE VISIT

Inputs. The material line of a performance restoration at this Part's midpoint, \$1.4 million × 60–70% = \$840,000 to \$980,000, taken at its midpoint of \$910,000 (sourced shares: Air Cargo Week, 2025/26; visit price sourced: Safe Fly Aviation, consultancy blog; midpoint assumed). Three forward

escalation readings over three years: about 10% a year (sourced: Aviation Week, late 2023, the 2022–23 observed rate for spares); about 7% a year (derived in Part I on life-limited-part sets); and Safran's rule of inflation plus three to four points, with inflation assumed at 3% for illustration, giving 6% to 7%.

at 10% a year: $910,000 \times 1.10^3 = 910,000 \times 1.331 = \$1,211,000$ (+\$301,000)
 at 7% a year: $910,000 \times 1.07^3 = 910,000 \times 1.225 = \$1,115,000$ (+\$205,000)
 at 6% to 7% a year: $910,000 \times 1.191$ to $1.225 = \$1,084,000$ to $\$1,115,000$
 (+\$174,000 to +\$205,000)

So over three years the material line alone rises by about \$170,000 to \$300,000 on a visit that today costs \$1.4 million, depending on which reading is used, and the increase lands on the same line whether the shop is the manufacturer's, an independent's or an airline's. On the alternative visit price of \$2.15 million (Safe Fly's other range) every dollar figure above scales by about 1.54. Limits of the example: the actual 2024–26 steps are unknown (D2 U1); the 10% rate was observed on all spares in 2022–23 and the 7% rate on life-limited parts over seventeen years, so neither is a forecast; the inflation figure in Safran's rule is assumed; and the material share is a secondary figure (R-014). [Air Cargo Week, 2025/26; Safe Fly Aviation, 2026; Aviation Week, late 2023; Part I worked example 1.3; reconciliation R-013, R-014, R-018]

28.2 TIME-AND-MATERIALS AND LONG-TERM AGREEMENTS

The second distinction in the manufacturer's model is who pays for a visit and when.

TIME-AND-MATERIALS (T&M)

Under a time-and-materials arrangement the operator pays for each shop visit as it occurs: parts at the catalogue list price and labour by the hour, with the workscope agreed visit by visit. The operator carries the cost risk (a heavier workscope costs more) and the volume risk (more removals cost more), and it chooses the shop and, within the shop manual, the source of the parts. It is the pay-per-visit model and the opposite of a long-term agreement. [D2 §2.3, §8]

LONG-TERM SERVICE AGREEMENT (LTSA); ALSO LONG-TERM AGREEMENT (LTA), RATE-PER-FLIGHT-HOUR (RPFH) OR FLIGHT-HOUR AGREEMENT

A long-term service agreement is a multi-year contract under which the operator pays a fixed amount per engine flight hour and the manufacturer, or a shop, commits to perform or pay for the engine's shop visits. The provider takes the volume and cost risk and in return controls the workscope, the parts sourcing and the choice of shop. Revenue under such a contract is recognised against an estimate of the contract's total profitability, so a change in that estimate flows through current-period profit: GE's 2025 segment profit was reduced by "an unfavorable change in estimated profitability of long-term service agreements, primarily from the estimated impact from tariffs". About 15% of the CFM56 fleet is under such agreements against about 50% of the LEAP fleet, with the LEAP targeted at 60–70% by 2030. The dossiers use three names for one concept; this primer uses LTSA and gives LTA and RPFH as aliases once, here. Part III §20 glosses the term at first mention. [D2 §2.3; D9 §2.5; D11 §2.3; GE Aerospace, Form 10-K FY2025; registry C-12]

Safran's description of the CFM56 aftermarket is that 85% of its revenue is spare parts and time-and-materials work, against about 40% for the LEAP, and that the LEAP fleet will reach 60% to 70% coverage by long-term agreements by 2030. [Safran via Aviation Week, "CFM Aftermarket Shifting To Long-term Agreements", c. 2023/24] The fleet-share figures are complements of the same picture: about 15% of CFM56 engines under long-term agreements against about 50% of the roughly 5,500 LEAPs in service at the time of the report. [Aviation Week, "Leap Aftermarket Poised For Growth", undated; reconciliation B-4] Both figures are Aviation Week's reporting of Safran's view, so their agreement is weak as corroboration. [reconciliation B-4]

The distinction matters to every later Part of this primer for one reason. Under a time-and-materials visit the operator, or the shop it chooses, decides whether a part is new from the manufacturer, used, repaired or an approved alternative; under a long-term agreement the manufacturer decides, and it has no reason to choose a competitor's part. [D2 §2.3] So the used-material market of Part V, the alternative parts of Part VI and the module exchanges of Part VII all address the 85% of the CFM56 aftermarket that is transactional, and would address far less of a fleet that was contracted the way the LEAP is being contracted. The research found no source that quantifies how much CFM56 parts demand sits inside long-term agreements and so outside the open market. [D2 §3.4]

28.3 TRUECHOICE: THE FOUR OFFERINGS

Part I §9 boxed TrueChoice as GE Aerospace's branded menu of CFM56 and LEAP services. This subsection gives each of its four offerings in GE's own words and says what each covers, because one of them is the manufacturer's direct equivalent of the product FTAI sells.

EXHIBIT 4.8 — GE AEROSPACE'S TRUECHOICE MENU: THE FOUR OFFERINGS, WHAT EACH COVERS, AND THE REFERENCE CUSTOMERS THE RESEARCH REACHED

Offering	GE's description	What it covers, functionally	Reference customers reached	Where it sits in this primer
TrueChoice Flight Hour	"Customized offerings that help optimize cost of ownership over the entire engine lifecycle with flexible risk transfer and payment options"	A rate-per-flight-hour agreement: the manufacturer takes cost and volume risk and controls workscope and parts	Southwest Airlines, CFM56-7B fleet (agreement expanded)	The long-term agreement of §28.2
TrueChoice Overhaul	"Time and material overhauls with tailored worksopes specific to shop visit objectives, economic priorities and ownership horizon — whether for one engine or an entire fleet"	Pay-per-visit shop visits performed in GE's own network	Safair and TAAG (Angola's flag carrier), CFM56	A time-and-materials visit in an OEM shop
TrueChoice Material	"High-quality new and used OEM parts, advanced repairs and technology upgrades for airlines and MROs that enhance engine performance over the lifecycle and support higher engine residual value"	New and used manufacturer parts, repairs and upgrades sold to third-party shops and to airlines using them; where the manufacturer's used-material business sits	Ryanair, long-term materials agreement, about 2,000 CFM56 and LEAP engines (GE 10-K FY2025)	The parts slice of §29; the used-material competitor of Part V
TrueChoice Transitions	"A broad range of options for changing ownership horizons, including green time leases, exchanges and material buy-back, plus custom worksopes with shorter builds and maximum used material"	Green-time leases, module and engine exchanges, buy-back of material, and shorter, used-material-heavy builds for operators that will not keep the engine long	None named in the research	The direct analogue of module exchange (Part VII §46) and of the green-time trade (Part II §16)

Source: GE Aerospace press releases on Safair, TAAG and Southwest Airlines TrueChoice agreements (undated); GE Aerospace, Form 10-K FY2025 (Ryanair); MRO Global on the Safair agreement. Compiled in D2 §2.3. The quotation marks carry GE's wording, including its own punctuation.

Read down the third column and the menu maps onto the rest of this Part. Flight Hour is the long-term agreement. Overhaul is a time-and-materials visit that happens to be performed in the manufacturer's shop. Material is the parts slice sold to everyone else, and it is where GE's statement that "GE and CFM are the largest used serviceable material provider for their products" belongs. [GE Aerospace investor-relations page, 2026] Transitions is the one that matters most to Part VII: green-time leases, exchanges, material buy-back and "custom worksopes with shorter builds and maximum used material" are, in the manufacturer's own words, the products an operator buys when it does not intend to keep the engine, and they are the same products FTAI's Module Factory and the independents' end-of-life offerings sell. [GE Aerospace press releases; D2 §2.3] The research found no price, volume or customer for Transitions, so the analogy is structural and not quantified. [D2 §2.3, U7]

28.4 MATERIAL SOLUTIONS, AND THE MANUFACTURER AS A SELLER OF USED PARTS

MATERIAL SOLUTIONS

Material solutions is the generic trade phrase for a shop's or a manufacturer's package of used serviceable material, repaired parts and approved alternative parts assembled to reduce the new-parts bill of a shop visit. The whole commercial logic of a shop that is not the manufacturer sits on this line, since material is 60–70% of a visit. The research could not match the phrase to a current CFM or GE programme name; GE's used-material offering sits inside TrueChoice Material, and a historical Pratt & Whitney "Global Material Solutions" venture on the CFM56-3 was found only as a headline. This primer uses the generic sense. [D2 §2.2, §8; D4 U10; registry C-19]

The manufacturer competes on this line too. GE states that "GE and CFM are the largest used serviceable material provider for their products", and that it will "continue to strengthen MRO access to OEM materials to support further CFM56 longevity". [GE Aerospace investor-relations page, 2026] The name, scale and CFM56 role of GE's used-material arm were not captured, and the research records the gap as a cluster shared with Parts V and VI. [D3 U15; D4 U10; cluster K-16] What the pricing of used material against new looks like, 60% to 80% of a new part's price on average or 70% to 75% of list when a trader sells to an airline, is Part V §34's subject, and the two figures are carried there as a pair. [Aviation Week, undated; Aircraft Commerce Issue 120; reconciliation R-022]

28.5 REPAIR MANAGEMENT AND OVERHAUL CONSULTING

REPAIR MANAGEMENT

Repair management is a service the manufacturer sells to a shop that is not its own: it manages the routing of the shop's piece-part repairs through the manufacturer's repair network and data, so the independent shop buys repair capability rather than building it. GE sold such a contract to Aero Norway, a CFM56 independent. A related product is overhaul consulting, under which the manufacturer advises an airline's own shop on its overhauls; EgyptAir signed a TrueChoice overhaul-consulting agreement with GE. The research holds the names of the two contracts and no terms. [GE Aerospace release, "Aero Norway awards repair management contract to GE Aviation", undated; Amwal Al Ghad, undated; D2 §2.3]

These two products show the manufacturer earning from visits it does not perform. A repair-management contract puts the manufacturer's repairs into an independent's visit; an overhaul-consulting agreement puts its workscope judgement into an airline's. Both leave the shop independent in name and bring part of the visit's repair slice (§29.2) back to the manufacturer. [D2 §2.3]

28.6 IAE ENGINEWISE AND FLEETCARE

Part I §10 introduced the V2500 and boxed EngineWise as Pratt & Whitney's aftermarket services brand, under which the IAE network performs V2500 shop visits. IAE is the consortium of Pratt & Whitney, part of RTX (formerly Raytheon Technologies), Pratt & Whitney Aero Engines International, Japanese Aero Engines Corporation and MTU Aero Engines. [D2 §2.3; Part I §10] RTX describes EngineWise as "a variety of aftermarket services to maximize engine performance and fleet availability". [RTX / Pratt & Whitney release, 6 Jun 2024]

The agreement that brings EngineWise into this primer is FTAI's. On 6 June 2024 IAE and FTAI announced a five-year EngineWise agreement covering more than 100 full performance-restoration shop visits on V2500 engines, which RTX called "one of IAE's largest engine maintenance agreements by number of engines with a non-airline customer"; FTAI then owned or was committed to more than 140 V2500 engines and aimed for 200 by 2025. [RTX / Pratt & Whitney release, 6 Jun 2024] Functionally this is the opposite of the Module Factory: a large engine owner buying manufacturer-network restorations at a negotiated price for a committed volume. Part VII §51 carries it. [D5 §2.5]

The V2500 market the agreement sits in has two counts for 2025 and the primer carries both. Aviation Week's V2500 data tool projected a spike to about 1,400 V2500 shop visits in 2025, with more than 7,800 overhaul visits and 3,142 life-limited-part-driven visits over the following ten years. [Aviation Week, "Data Tool: The Future Of IAE V2500 Engine", 2024/25] RTX's chief financial

officer, as reported by Aviation Week in 2026, put 2025 V2500 shop visits at "slightly more than 800", with 2026 expected within about 20 of that. [RTX CFO via Aviation Week, 2026; reconciliation R-009] One is a published forecast and the other a reported outturn, both via the same publication, and the research does not resolve them. IAE separately said it expects no V2500 retirement surge over the next five years. [IAE via Aviation Week, 2024/25]

"FleetCare" was not found as a current IAE or Pratt & Whitney product name; only EngineWise was. [D2 U9] The brief for this primer named it, and the primer records that the name did not surface rather than describing a product it did not find.

28.7 THE MANUFACTURERS' REPORTED NUMBERS

EXHIBIT 4.9 — THE OEM AFTERMARKET IN THE MANUFACTURERS' OWN FIGURES, 2023 TO 1Q 2026

Item	Figure	Source	Date
GE CES revenue, 2025	\$33,314M; equipment \$8,304M; services \$25,010M (75.1%)	GE Aerospace 10-K FY2025	Feb 2026
GE CES segment profit, 2025	\$8,861M; margin 26.6%; up \$1.8B (26%)	Same	Feb 2026
GE CES revenue growth, 2025	Up \$6.4B (24%): "increased spare parts volume, internal shop visit volume and worksopes, increased engine deliveries and pricing"	Same	Feb 2026
GE internal shop-visit revenue growth	+24% (2025); +19% (2024); +27% (2023)	Same	Feb 2026
GE CES services revenue growth, 2025	+26%	Same	Feb 2026
GE commercial engine deliveries	2,386 (2025) vs 1,911 (2024); LEAP 1,802 vs 1,407	Same	Feb 2026
GE services revenue, 1Q 2026	"Services revenue grew 39%"	GE Aerospace 1Q 2026 release	Apr 2026
GE external spares, late 2023	+35% year on year; \$42.2M a day shipped	Aviation Week	Late 2023
GE statement on LEAP	"Expected in the coming years to overtake the mature CFM56 as the industry's largest fleet", driving "a significant increase in shop visits and need for MRO capacity"	GE Aerospace 10-K FY2025	Feb 2026

Item	Figure	Source	Date
CFM56 share of GE CES services	Not disclosed	D2 U4; D11 U16	n/a
Safran group revenue, 2025	€31.3B (+14.7%); recurring operating income €5.2B (16.6%)	Safran release	13 Feb 2026
Safran civil spares (USD), 2025	+17.6%, "driven primarily by the CFM56"	Same	13 Feb 2026
Safran Propulsion services	€10,122M, 64.6% of Propulsion revenue (2024: €8,430M, 61.7%)	Safran FY2025 results presentation	13 Feb 2026
Safran 2026 guidance	Revenue growth low-to-mid teens; recurring operating income €6.1–6.2B	Safran release	13 Feb 2026
Safran civil aftermarket guidance, 2023	Raised twice to "low-30% range" growth	Aviation Week	Late 2023
CFM56 aftermarket mix	85% spares and time-and-materials (LEAP about 40%)	Safran via Aviation Week	c. 2023/24
Long-term agreement coverage	About 15% of CFM56 fleet; about 50% of LEAP fleet; LEAP target 60–70% by 2030	Aviation Week; Safran via Aviation Week	Undated; c. 2023/24

Source: GE Aerospace, Form 10-K FY2025; GE Aerospace, 1Q 2026 earnings release; Aviation Week, "Parts Price Hikes Help Boost Safran, GE Aftermarket Sales", late 2023; Safran, "Safran reports excellent financial performance in 2025", 13 Feb 2026, and FY2025 Results & Investor Update presentation; Forecast International and Leeham News, 13 Feb 2026; Aviation Week, "CFM Aftermarket Shifting To Long-term Agreements" and "Leap Aftermarket Poised For Growth". Compiled in D2 §4.3 and D11 §3.8–3.9. Unknowns U-D2-04, U-D11-16.

The exhibit is the manufacturers' view of the market the shops of §25 work in. GE's services revenue of \$25 billion in 2025 is the size of the pool that FTAI's Aerospace Products segment, at \$875.0 million of revenue in the second quarter of 2026, draws from, and the two are not comparable beyond that sentence (Part XI §89). [GE Aerospace, Form 10-K FY2025; FTAI release, 29 Jul 2026; D11 §3.8] Three things the exhibit does not contain are worth naming. No CFM56 share of GE's services revenue is disclosed, although GE has historically described the CFM56 as its largest services contributor. [D2 U4] No CFM56 shop-visit count, spare-parts price percentage or used-material programme wording was returned from GE's filing. [D11 U16] And Safran's US-dollar spare-parts growth figure for the CFM56 alone was not returned. [D11 U17]

28.8 THE CFM–FTAI MATERIALS AGREEMENT OF JANUARY 2026

On 22 January 2026 FTAI announced a multi-year materials agreement with CFM International under which FTAI "secures OEM replacement part supply, thrust performance upgrades and component repair", described by both companies as strengthening "the open maintenance, repair, and overhaul (MRO) ecosystem". [FTAI release, 22 Jan 2026] The agreement places FTAI, a buyer of used material and a partner in an alternative-parts venture, on the manufacturer's TrueChoice Material side of the ledger at the same time; on FTAI's fourth-quarter 2025 call its chief executive named it, with the approved turbine blade and more used material, as a lever for moving segment margins from the mid-30s toward 40%. [MarketBeat coverage of the FTAI call, 26 Feb 2026] Its duration, volumes, pricing and any effect on FTAI's use of alternative parts were not retrieved, and the release body itself was located and not read. [D4 U14; D5 U18; cluster K-19] Part VII §52 carries the agreement within the segment.

29. Where the money goes in a shop visit, and who captures it

Part I §7 built the cost of a shop visit from five lines: labour, new parts, repairs, life-limited parts and the test-cell run. This section asks who receives each line. The answer is what makes the manufacturer's model of §28 bite on every visit, and it is the frame for Parts V, VI and VII, each of which describes one way of moving money off the manufacturer's line.

29.1 THE SHARES

Two share breakdowns exist and they come from different secondary sources. Air Cargo Week puts material at 60% to 70% of a visit, labour at 20% to 30%, and repairs and other shop activities at 10% to 20%. [Air Cargo Week, "The true cost of engine maintenance", 2025/26] Safe Fly Aviation, a consultancy blog, gives a finer cut for the CFM56: life-limited parts are 40% to 60% of the parts cost when they are replaced, high-pressure turbine blades and vanes are 15% to 25%, and labour is 15% to 20%. [Safe Fly Aviation, 2026] Neither source names a shop-level data set, and the labour shares differ. [reconciliation R-014] One dossier reads the two as consistent, since life-limited parts and turbine airfoils together are the bulk of material and material is the bulk of the visit; the primer reports both and does not combine them. [D2 §2.2]

29.2 SLICE BY SLICE: WHO RECEIVES IT

The exhibit below is the research's own assembly of who captures each slice, from the sources named in each row; it is the input to the margin pool that Part I §11 previewed and Part XI §86 treats in full. [D2 §2.2; registry 1c]

REPAIR VENDORS; COMPONENT REPAIRS

Component repairs are the restoration of parts that can be repaired rather than scrapped, such as turbine airfoils, cases and seals, by welding, re-coating, machining or re-plating. A repair vendor is the specialist shop to which an engine shop subcontracts that work: either the manufacturer's own repair shops, working to the shop manual, or independent specialists holding repairs approved through a Designated Engineering Representative (DER), treated in Part VI §40. Repairs are 10–20% of a visit, and a repaired part costs a fraction of a new one. [D2 §2.2; Air Cargo Week, 2025/26]

EXHIBIT 4.10 — WHERE A SHOP VISIT'S MONEY GOES AND WHO CAPTURES EACH SLICE

Slice	Share of the visit	Who captures it	Mechanism	Where the primer treats it
New OEM parts other than LLPs: blades, vanes, seals, hardware	The largest single slice within material's 60–70%	CFM International, that is GE Aerospace and Safran, at the catalogue list price	Sold through CFM's spares organisation to any shop; the shop passes it through at list plus a handling margin, or the airline buys direct under a TrueChoice Material agreement	§28.1, §28.3
Life-limited parts	40–60% of parts cost when replaced	The manufacturer only, in practice: no approved alternative life-limited part exists for the CFM56	Sold at list; the most visible escalator (the "\$4M stack" headline)	Part I §4; §30
Used serviceable material	Substitutes for new parts	Teardown and used-parts traders (AerSale, GA Telesis, others; lessors' part-outs) and the manufacturer, which calls itself the largest used-material provider for its products	Priced as a discount to list: 60–80% of new on average (Aviation Week); 70–75% of list from a trader to an airline (Aircraft Commerce); above list when the manufacturer cannot deliver	Part V §34
Approved alternative (PMA) parts	Alternative to new OEM parts	PMA makers: HEICO (a maker of approved alternative parts); Chromalloy, with FTAI as its partner	Priced at a discount to list that no retrieved source discloses	Part VI

Slice	Share of the visit	Who captures it	Mechanism	Where the primer treats it
Component repairs	10–20%	Repair vendors: the manufacturer's repair shops (its repair-management offering) and independents holding DER repairs	The shop subcontracts airfoil, case and seal repairs; a repaired part costs a fraction of a new one	§28.5; Part VI §40
Labour and test	20–30% (Air Cargo Week) or 15–20% (Safe Fly)	The engine shop, whoever owns it	Shop rate × hours; labour-rate inflation 5.5–6.0% in 2025 (Oliver Wyman)	§27.3
Shop margin and risk	Embedded in the lines above	The shop under time-and-materials; the manufacturer under a flight-hour agreement	Under T&M the operator bears cost and volume risk; under an LTSA the manufacturer bears it and prices it in	§28.2

Source: Air Cargo Week, "The true cost of engine maintenance", 2025/26; Safe Fly Aviation, 2026 (consultancy blog); GE Aerospace investor-relations page, 2026; Aviation Week, "MRO Memo: A Seller's Market For Used Parts"; Aircraft Commerce Issue 120; Oliver Wyman, Apr 2026; D4 on the absence of PMA life-limited parts. Assembled in D2 §2.2. Reconciliation R-014, R-022.

Three of the rows need a name the reader has not yet met in this Part.

CHROMALLOY

Chromalloy is a Florida-based maker of turbine-engine part repairs, coatings and approved alternative parts, with revenue described as "\$1 billion-plus" and about 4,000 employees, owned by Sequa Corporation (a Veritas Capital affiliate since December 2022). In October 2025 it received approval from the FAA (the US Federal Aviation Administration) for a CFM56-5B/7B high-pressure turbine blade, which it calls "the only aftermarket alternative to the OEM manufactured new part". Part VI §41 carries the company. [Chromalloy release, 30 Oct 2025; D4 §3]

FTAI-CHROMALLOY JOINT VENTURE

FTAI's annual report says the company "develops and manufactures PMA parts through a joint venture"; the partner, Chromalloy, is named in Chromalloy's own releases and in trade coverage, and investor commentary describes the arrangement as an exclusive partnership formed in 2018 covering several CFM56 hot-section parts. Whether the "Advanced Engine Repair JV" in FTAI's equity-method note (25%, carrying value \$22.4 million at 31 December 2025) is this venture is unknown. Part VI §43 carries what is filed and what is reported. [FTAI 10-K FY2025; Chromalloy release, 30 Oct 2025; orchestrator notes; reconciliation R-026]

HEICO FLIGHT SUPPORT GROUP

The Flight Support Group is the segment of HEICO (a Florida-based maker of approved alternative parts) Corporation that designs and sells approved alternative parts and repairs for commercial aircraft and engines; it reported fiscal 2025 net sales of \$3,117.3 million and operating income of \$750.4 million (24.1%), and HEICO adds several hundred new part approvals a year, read as 300–500 by one dossier and 400–550 by another from the same filing. Part VI §41 carries it. [HEICO, Form 10-K FY2025; reconciliation R-024]

Two rows in the exhibit are the whole point of the manufacturer's model. The life-limited-part row has one supplier, because the research found approved alternative turbine blades and vanes but no alternative disk or spool for the CFM56, so every new life-limited part is a single-source purchase at the escalating list price (Part I §4, Part VI §38). [D4 U7; D1 §5] The dossier that assembled the exhibit calls the full set of life-limited parts the "LLP stack"; this primer uses Part I's term, the full LLP set, and gives the alias once here. [registry, Part IV rows] The used-material row has the manufacturer on both sides: it sells the new part at list and, by its own statement, is the largest seller of the used part priced as a discount to that list. [GE Aerospace investor-relations page, 2026]

29.3 THE TURBINE-BLADE LINE: A TENSION AT THE LEVEL OF ONE PART

The high-pressure turbine's blades and vanes are the most expensive non-life-limited parts in a visit, and the research holds two readings of what they cost that do not agree. [reconciliation R-015]

On one side is the share. Blades and vanes are 15% to 25% of a visit's parts cost, which on this Part's restoration range of \$1.2–1.6 million is roughly \$0.18–0.40 million and on Part I's range of \$1.8–2.5 million roughly \$0.27–0.63 million. [Safe Fly Aviation, 2026; arithmetic in the research, D1 §2.7 and D2 §2.2] On the other side is the price per blade. An expert interview puts a new manufacturer's CFM56 high-pressure turbine blade at about \$20,000; a web listing of weak provenance, whose condition is unclear, shows about €35,000. [In Practise, undated; web listing via D4] At a commonly cited but unverified 80 blades per rotor, a set is \$1.6 million at the interview price or about \$3.08 million at the listing price, and the same interview says a new set of blades and vanes "can cost up to a few million dollars". [D4 §2.8; In Practise, undated; D4 U16]

The two readings cannot both describe the same visit. A blade set alone at \$1.6 million is more than the whole restoration on this Part's range. The research did not retrieve the manufacturer's list price for a full blade-and-vane set or confirm the blade count, so the primer carries both sides and stops. [reconciliation R-015; D4 U16] The interview adds a cadence that bears on which visits the line applies to: at a first shop visit, around 20,000 cycles, airlines typically repair the turbine blades; at a second, around 30,000, hot-section parts typically cannot be repaired again and new parts are needed, which is where an approved alternative blade competes with a new manufacturer's blade. [In Practise, undated, via D4 §2.8] Part VI §44 works the parts bill with and without the alternative part on both readings.

29.4 WORKED EXAMPLE: ONE RESTORATION, SPLIT BY SLICE AND BY PARTY

WORKED EXAMPLE 4.4 — A CFM56-7B PERFORMANCE RESTORATION SPLIT BY SLICE AND BY WHO RECEIVES EACH SLICE, ON BOTH COST SETS

Inputs. Two visit prices: \$1.4 million, the midpoint of this Part's \$1.2–1.6 million range (sourced: Safe Fly Aviation, consultancy blog, "2026 industry data"), and \$2.15 million, the midpoint of Part I's \$1.8–2.5 million range (sourced: Safe Fly Aviation, other page). Shares of material 60–70%, labour 20–30% and repairs 10–20% (sourced: Air Cargo Week, 2025/26); turbine blades and vanes at 15–25% of parts cost and labour at 15–20% (sourced: Safe Fly Aviation). No life-limited parts are replaced in this visit, so the 40–60% LLP share does not apply. The three Air Cargo Week ranges do not sum to 100% at their ends, so the lines below are each range applied to the total and are not additive.

on \$1.4M: material 840,000–980,000; labour 280,000–420,000 (Safe Fly share: 210,000–280,000); repairs 140,000–280,000
 turbine blades and vanes within material: $15\text{--}25\% \times 840,000\text{--}980,000 = 126,000\text{--}245,000$
 on \$2.15M: material 1,290,000–1,505,000; labour 430,000–645,000 (Safe Fly share: 323,000–430,000); repairs 215,000–430,000
 turbine blades and vanes within material: 194,000–376,000

Who receives each line. The material line goes to the manufacturer at list for every part bought new, so on an all-new-parts visit the manufacturer receives \$840,000 to \$980,000 of a \$1.4 million visit, or \$1.29 to \$1.51 million of a \$2.15 million one, whichever shop performs it. The labour line stays with the shop. The repairs line goes to repair vendors, which may be the manufacturer's repair shops or independents holding their own approved repairs. Now suppose, for illustration only, that the shop sources 20% of the material line's list value as used parts bought at 70% of list (assumed share; discount sourced as the lower end of Aircraft Commerce's 70–75% trader price):

material at midpoint of the \$1.4M visit: 910,000
 list value sourced used: $20\% \times 910,000 = 182,000$; paid to the trader at
 70%: 127,400
 manufacturer receives: $910,000 - 182,000 = 728,000$; shop saves: $182,000 - 127,400 = 54,600$ (3.9% of the visit)

So each fifth of the material line moved from new to used at a 30% discount saves about 4% of the visit's price and moves about \$180,000 of list value from the manufacturer to a trader; an approved alternative part or a repair in place of a replacement moves money the same way, at discounts no retrieved source discloses. Limits of the example: the two visit prices are a contested pair (R-013); the shares are secondary and disagree with each other (R-014); the turbine-blade line is contested at the level of one part (R-015); the 20% used-material share is assumed; and the shop's handling margin on parts passed through at list, which is not zero, is not separated out because no figure was found. [Safe Fly Aviation, 2026; Air Cargo Week, 2025/26; Aircraft Commerce Issue 120; D2 §2.2; reconciliation R-013, R-014, R-015, R-022]

29.5 WHAT FOLLOWS FROM THE SPLIT

Two consequences follow from the exhibit and the example, and the research states both. First, because 60% to 70% of a visit is material and most material is priced by the manufacturer, the manufacturer's annual catalogue escalation flows almost directly into the cost of every shop visit regardless of who performs it; worked example 4.2 put the figure at roughly four to seven points of visit cost a year on the two escalation readings. [D2 §2.2; Air Cargo Week, 2025/26] Second, the only levers a shop other than the manufacturer has to cut the bill are on the material line: used parts instead of new, approved alternative parts instead of the manufacturer's, repair instead of replacement, and module exchange instead of overhaul. [D2 §2.2] Parts V, VI and VII take those four levers in turn, and §30 sets the fourth against the other three in one example.

WHY THE MANUFACTURER'S CAPTURE ON THE CFM56 IS PARTS, NOT CONTRACTS

Put §28.2 and Exhibit 4.10 together. About 85% of the CFM56's aftermarket revenue is spare parts and time-and-materials work, so most visits are chosen and paid for visit by visit by the operator or its shop. Within each of those visits, 60% to 70% of the bill is material, and most material is bought new from the manufacturer at list. So the manufacturer receives the largest slice of nearly every CFM56 visit without performing it, and without a contract that obliges the operator to come back. On the LEAP the structure is reversed: about half the fleet is already under long-term agreements and the target is 60% to 70%, so the manufacturer's capture there moves from the parts line to the contract, under which it also chooses the parts. The CFM56's open structure is what leaves room for the material-line substitutes of Parts V to VII; the LEAP's contracted structure is what Part III §24's rows on LEAP durability and the GTF groundings say does not create demand for those substitutes. [Safran via Aviation Week, c. 2023/24; Aviation Week, undated; Air Cargo Week, 2025/26; D2 §2.3, §3.4; D9 §5]

30. Worked example: four options for an engine at its LLP limit

Everything in this Part meets in one decision. An engine on a mid-life narrowbody is approaching the limit of a life-limited part in its core. The owner can send it to a shop for a full visit, swap the module that has run out, lease a part-worn engine to fly in its place, or sell it and buy another. Each option draws on a different part of the market described above: the first on the shops and slots of §25 and §27 and the parts prices of §28, the second on the module exchange that Part VII sells and GE packages as TrueChoice Transitions, the third and fourth on the green-time and used-engine trade of Part II §16 and Part V. The example below prices all four on the research's figures, with every input labelled sourced or assumed and every contested figure shown with its alternative. It is arithmetic, not advice.

Two aliases from the research need a word before the arithmetic. The dossier behind this section calls the full set of life-limited parts the "LLP stack" and calls the mismatch of remaining lives across those parts "LLP stagger"; Part I §4 treats the same two things as the full LLP set and as stub life, and this section uses Part I's terms with the aliases given once here. [D2 §2.5, §8; registry, Part IV first-use rows] Part I §4 also boxed cycles since new (CSN), which the exhibit below uses, as the cycles an engine or part has flown since manufacture.

WORKED EXAMPLE 4.5 — FOUR OPTIONS FOR A CFM56-7B WHOSE HIGH-PRESSURE TURBINE DISK IS 1,200 CYCLES FROM ITS LIMIT

Setup. An airline operates a 737-800 with two CFM56-7Bs. On one engine the high-pressure turbine disk, a life-limited part with a certified limit of 20,000 cycles (sourced: records mini-packs, Part I Exhibit 1.5) and a new price of \$420,000 (sourced: Safe Fly Aviation, consultancy blog, 2026), has 1,200 cycles remaining (assumed). The engine's temperature margin would otherwise allow about 3,000 more cycles (assumed). The aircraft flies about 1,500 cycles a year (sourced as the upper end of Aircraft Value News's 1,000–1,500 cycles a year for a mature narrowbody). The airline plans to retire the aircraft in 24 months (assumed), so it needs about 3,000 cycles from this engine position. Because the limiting part is in the turbine, any action on it forces the core open (Part I §4).

The per-cycle frame. A full set of -7B life-limited parts is about \$4 million at list (sourced: Aircraft Value News headline, Nov 2019; this Part's figure under E.1 rule 5) or \$5.7 million (sourced: MyAirTrade engine-status table, June 2025, scope not visible; Part I's figure). Over a 20,000-cycle life:

at \$4.0M: $4,000,000 \div 20,000 = \200 per cycle; at 1,500 cycles a year,
\$300,000 per engine per year
at \$5.7M: $5,700,000 \div 20,000 = \285 per cycle; \$427,500 per engine per
year

That cost accrues whether or not the engine ever goes to a shop, which is why lessors collect reserves against it (Part II §13) and why an engine's value is so sensitive to its remaining life (Part I §8).

Option A: a full shop visit with life-limited-part replacement. The restoration workscope is \$1.2–1.6 million, midpoint \$1.4 million (sourced: Safe Fly Aviation, "2026 industry data"; this Part's range under E.1 rule 4); the alternative range from the same publisher is \$1.8–2.5 million, midpoint \$2.15 million (Part I's range). A full new set of life-limited parts is \$4.0 million, or \$5.7 million on the alternative.

$$\begin{aligned} \text{outlay, full set: } & 1,400,000 + 4,000,000 = \$5,400,000 \\ \text{alternatives: } & 1,400,000 + 5,700,000 = \$7,100,000; 2,150,000 + \\ & 4,000,000 = \$6,150,000; 2,150,000 + 5,700,000 = \$7,850,000 \end{aligned}$$

Variant A' replaces only the turbine's life-limited parts, "easily \$0.5m" at 2017 list (sourced: Leeham News, 3 Mar 2017; a 2017 figure, so understated at 2026 prices): \$1.4 million + \$0.5 million = \$1.9 million (or \$2.65 million on the alternative restoration). The other life-limited parts then keep their own, lower, remaining lives, so the next life-limit-forced removal comes sooner; this is the stub-life problem of Part I §4, which the dossier calls LLP stagger. Downtime is the slot wait of two to six months (sourced: Bain, 2024) plus the turnaround, 75–90 days (sourced: Safe Fly) or 90–120 days (sourced: trade press, 2025) or about 90 days (sourced: GE, May 2026). Covering the position with a leased spare for three months of turnaround alone costs \$126,000–144,000 at \$42,000–48,000 a month (sourced: Safe Fly, 2026, a rate that sits beside green-time values and may describe a green-time engine) or about \$300,000 at about \$100,000 a month (sourced: IBA, Apr 2024); the dossier's figure of about \$135,000 is the first of these. Adding a slot wait of two to six months takes the cover to \$210,000–432,000 at the lower rate or \$500,000–900,000 at the higher. The value effect: a low-life -7B trades as teardown feedstock at \$0.8–1.2 million and a green-time -7B at \$2.8–3.4 million (both sourced: Safe Fly, 2026, consultancy blog); IBA's half-life market values of \$5.7 million for a -7B24 and \$6.4 million for a -7B27 (sourced: IBA, Sept 2025) are a reference for an engine with life, on a different basis. What a freshly restored engine with a full new set is worth was not found (D2 U8), so the dossier takes the green-time midpoint as a floor:

$$\begin{aligned} \text{value uplift, floor: } & 3,100,000 - 1,000,000 = \$2,100,000 \\ \text{net economic cost over the horizon, engine sold at exit: } & 5,400,000 + \\ & 135,000 - 2,100,000 \approx \$3,400,000 \\ \text{on the alternative restoration ($2.15M): } & \approx \$4,200,000; \text{ on the} \\ \text{alternative set ($5.7M): } & \approx \$5,100,000; \text{ on both: } \approx \$5,900,000 \end{aligned}$$

The true figure is lower if the fresh engine fetches more than a green-time price, and the airline has bought about 20,000 cycles of life to use 3,000 of them, recovering the other 17,000 only through the resale price.

Option B: a module exchange. The airline swaps the turbine module, or the whole core, for a serviceable one from a provider's inventory carrying, say, 10,000 cycles of remaining life (assumed); the fan and low-pressure turbine stay. The engine is opened only to the core split, so turnaround is

shorter and labour is lower than in Option A. The price structure is an exchange fee plus the value of the life transferred, adjusted for the condition of the other hardware (Part I §3; Part VII §47). The life transferred, if priced at the stack-average new-list rate: $10,000 \times \$200 = \2.0 million (or $10,000 \times \$285 = \2.85 million on the alternative set). For the turbine's own life-limited parts alone, \$0.5 million over 20,000 cycles is \$25 a cycle, so 10,000 cycles is about \$250,000. Safe Fly quotes "removal costs of \$150,000–\$400,000 per module" for life-limited-part modules, whose meaning is unclear (D2 U7). The exchange fee itself, what FTAI, GE under TrueChoice Transitions or MTU charge above the life transferred, was not found in any source, and the research records it as a cluster shared with Parts V and VII (D2 U7; cluster K-01). The arithmetic therefore brackets Option B:

turbine-set exchange: [fee, unknown] + ~250,000 $\approx$ \$500,000 to \$1,000,000
 (the dossier's bracket)
 full-core exchange: higher, up to [fee] + ~2,000,000 if a whole stack's
 life is priced at new list

Downtime for a quick-turn of about 45 days (sourced: trade press, 2025) costs about \$63,000–72,000 of cover at the lower rent or about \$150,000 at the higher. The only FTAI figures the research holds are bounds, not prices: products revenue per module produced of \$2.11 million in 2025 and \$2.34 million in the second quarter of 2026, with the limits Part VII §56 lists (E.1 rule 11). Part VII §55 carries FTAI's own exchange-against-visit example.

Option C: a green-time engine lease. The airline removes the engine and installs a leased green-time -7B, typically carrying 4,000–8,000 cycles (sourced: Safe Fly), enough for the 3,000 it needs. Rent for 24 months:

at \$42,000–48,000 a month: $24 \times 45,000 = \$1,080,000$ (range 1,008,000–1,152,000)
 at about \$100,000 a month: $24 \times 100,000 = \$2,400,000$ (IBA's rate may describe a full-life spare, not a green-time engine)
 sell the removed engine as feedstock: +\$800,000 to +\$1,200,000
 net at the lower rent: $-1,080,000 + 1,000,000 \approx -\$80,000$ (range -\$352,000 to +\$192,000)
 net at the higher rent: $\approx -\$1,200,000$ to $-\$1,600,000$

Both nets are before any per-hour or per-cycle reserves the lessor charges and before redelivery conditions; no sourced reserve rate was found, and the only figures in the research are one dossier's illustrative assumptions (D6 U7). IBA's April 2024 commentary that "it's a lessor's market" and that engine lease rates and values were escalating is the context for both rents.

Option D: sell the engine. The airline sells the life-limited engine at its teardown value of \$0.8–1.2 million and buys a green-time unit at \$2.8–3.4 million (both sourced: Safe Fly): a net outlay of \$1.6–2.6 million, after which it owns an engine with 4,000–8,000 cycles, resaleable after 3,000 at a lower price the research did not find. Or it sells the whole aircraft as it stands and lets the buyer price the deficit: Safe Fly says two otherwise identical -7Bs at 10,000 cycles since new "differ by \$1.6 million

based solely on LLP remaining life", and that 60–80% of a used engine's value is its remaining life-limited-part life (consultancy blog).

Limits of the example. Every cost figure is from a consultancy blog or the trade press and no shop's price was read (R-013); the two restoration ranges and the two set prices are never combined within one line, and each line names its alternative (R-013, R-016, R-017); the two spare-engine rents may price different engines and are carried as a pair (R-028); the value figures sit on three bases (R-029); the exchange fee is a bracket with no source (U7); the reserve rates in Option C are unknown; and the value of a freshly restored engine is unknown (U8). The 2017 figure for the turbine's life-limited parts is understated at 2026 prices by an escalation whose steps for 2024–26 are themselves unknown (U1). [D2 §2.5; Safe Fly Aviation, 2026; Aircraft Value News, Nov 2019; MyAirTrade, June 2025; Leeham News, 3 Mar 2017; Bain, 2024; Aviation Business News, 2025; GE Aerospace, 27 May 2026; IBA, Apr 2024 and Sept 2025; Aircraft Value News, undated (cycles a year); reconciliation R-013, R-016, R-017, R-028, R-029; D2 U7, U8; D6 U7]

EXHIBIT 4.11 — THE FOUR OPTIONS SUMMARISED, ON BOTH SHOP-VISIT COST SETS, WITH THE SOURCES AND THE UNKNOWN IN EACH LINE

Option	Cash out over 24 months (this Part's cost set)	Cash out (Part I's cost set)	Cash or asset back	Indicative net	Downtime	Open items
A. Full visit, full new set	\$5.4M (restoration \$1.4M + set \$4.0M) + \$0.14M cover at the lower rent	\$6.15M (restoration \$2.15M + set \$4.0M); \$7.85M with the \$5.7M set	Engine worth at least the green-time \$3.1M midpoint; full-stack value unknown	About – \$3.4M, or better; –\$4.2M to –\$5.9M on the alternatives	Slot wait 2–6 months plus TAT 75–120 days	Fresh-engine value (U8); which rent applies (R-028)
A'. Restoration plus turbine LLPs only	\$1.9M + \$0.14M cover	\$2.65M + cover	Engine worth more than today, less than A	Not computable: next removal brought forward (stub life)	Same as A	2017 price for the turbine set; stub-life cost not priced

Option	Cash out over 24 months (this Part's cost set)	Cash out (Part I's cost set)	Cash or asset back	Indicative net	Downtime	Open items
B. Turbine or core module exchange	Fee (unknown) + about \$0.25M of life for the turbine set; up to fee + \$2.0M for a whole stack's life at new list	Same, since the restoration ranges do not enter	Engine with the life it needs	About –\$0.5M to –\$1.0M for a turbine-set exchange (dossier bracket)	Quick-turn, about 45 days; FTAI's own TAT not retrieved	Exchange fee (U7, cluster K-01)
C. Green-time lease, 24 months	\$1.0–1.15M rent at \$42–48k a month; \$2.4M at about \$100k	Same	+\$0.8–1.2M from selling own engine	About –\$0.1M (–\$0.35M to +\$0.19M) at the lower rent; about –\$1.2M to –\$1.6M at the higher; before reserves and redelivery	Engine swap only	Reserve rates (D6 U7); which rent (R-028)
D. Sell, buy a green-time engine	\$2.8–3.4M purchase	Same	+\$0.8–1.2M now; resale after 3,000 cycles later	–\$1.6M to –\$2.6M, less an unfound resale value	Engine swap only	Resale value after 3,000 cycles

Source: D2 §2.5, re-worked on both cost sets per reconciliation E.1 rules 4, 5 and 6; Safe Fly Aviation, "Engine Shop Visit Costs Worldwide 2026", "CFM56 Engine Market Report 2026", "CFM56-7B Engine Availability Report 2026" and LLP pages (consultancy blog); Aircraft Value News, Nov 2019; MyAirTrade, June 2025; Leeham News, 3 Mar 2017; Bain & Company, 2024; IBA via AviTrader and AJOT (a trade publication), 25 Apr 2024; IBA Engine Values 2025B, Sept 2025. CSN is cycles since new (Part I §4).

The arithmetic is not a recommendation, and the point is structural. Options B, C and D exist only because the engine is modular and because a secondary market in modules, green time and used material exists; without those, Option A would be the only option, and its price would be the manufacturer's price. [D2 §2.5] The three rows of Part III §24 that this Part answers meet in the table. Low retirements shrink the supply behind B, C and D, since fewer teardowns mean fewer modules, fewer green-time engines and fewer used parts, at exactly the moment the slot shortage

of §27 makes A slow and the escalation of §28 makes A expensive.^[D2 §2.5, §5.4] Which option an operator takes depends on inputs this Part has shown to be contested or unknown: the exchange fee, the lessor's reserve rates, the spare-engine rent, and the value of a restored engine against a part-worn one. Part V prices the used parts and the teardown, Part VI the alternative parts, and Part VII the exchange as FTAI sells it; Part II §16 carries the green-time trade.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-006	CFM56 shop visits a year: 2,300–2,400 in 2026–2028 (GE via Aviation Week, 2026) against about 2,500 peak in 2025–26 (Safran via Aviation Week, c. 2023/24); IBA "+40%" 2024 → 2025 headline (scope not captured) against Safran's "mid-single-digit" 2025 growth	§26.2, Exhibit 4.3
R-007	Peak shape and timing: 2,500 peak in 2025–26 (Safran) against a 2,300–2,400 plateau through 2028 with apex 2027–28 (GE); first-visit peak 2027 (Visual Approach)	§26.2, callout
D2 T6	Safran "topping 2,000 again in 2024" (including LEAP) against IBA's 40% increase 2024 → 2025; if one population, 2025 would be about 2,800	§26.2
R-004	Retirement rate: GE 3–4% → 2–3% → 1.5–2% (CFO via Aviation Week, 2026); 2025 about 1.5%, 2026 about 2% (via Leeham, 22 Jan 2026); 1Q 2026 "below 1%" (GE investor-relations page)	§26.3, Exhibit 4.4, worked example 4.1
R-005	Aircraft retirements 2024: "fewer than 200" (IBA, Sept 2024) against 587 (Cirium via Informa); scope not visible	§26.3, Exhibit 4.4
R-001	Installed base: about 24,000 (Aviation Business News, Sept 2025) against about 23,000 (CFM website, undated); both used in worked example 4.1	§26.3, worked example 4.1
R-008	Implied interval of about ten years (2,300 visits on 24,000 engines) against typical CFM56 intervals (Part I)	§26.3, worked example 4.1
R-012	Fleet maturity: 57% no visit and 21% one (GE, Jun 2019); about 70% zero or one (Safran, 2024); about 75% (Safe Fly, consultancy blog, 2026)	§26.4, Exhibit 4.4
R-037	FTAI capacity: "up to 300 shop visits per year" (10-K, 2022 language); 1,350 module overhauls (release, 9 Sep 2024); 1,800 modules / 600 engines (release, 26 Feb 2025); 3,000 modules (call coverage, Jul 2026)	§25.4, Exhibit 4.2
R-036	2026 module target: 1,050 (release, 25 Feb 2026) against 1,200 (call coverage, 30 Jul 2026)	§25.4
R-041	Lisbon: "planned 113,000 sq ft, 300+ modules" (call coverage, unverified) against "100% owned" and "still ramping" (10-K; call)	§25.4, Exhibit 4.2

Id	The two (or more) figures	Where in this Part
R-042	FTAI share: about 5% of a \$22B market (release, 26 Feb 2025); 25% goal of CFM56 shop visits, denominator not retrieved (call, Jul 2026); about 17% / 43% of a 2,350-visit year at 1,200 / 3,000 modules (derived)	\$25.4
R-019	Turnaround: 90–120 days full overhaul and about 45 light (trade press, 2025); "sustaining ~90 days" (GE, 27 May 2026); more than 10% better in 4Q 2025 (GE 10-K); 75–90 for a performance restoration (PR) and 120–150 heavy (Safe Fly); slot wait +2–3 months, up to 6 (Bain, 2024)	\$27.6, Exhibit 4.6
R-020 (D2 T8)	Parts constraint: priority-supplier input +40% and TAT +10% (GE 10-K FY2025) against labour and material shortages as "primary disruptors", two-thirds reporting next-generation shop costs 21%+ over expectation (Oliver Wyman, Apr 2026) and thin used-material supply (AVM, 2025)	\$27.4, Exhibit 4.5
R-018	Escalation: LLP set about 7% a year 2008–25, Ishka about 12% step (year not shown), -7B parts +20–30% (2023) against spares about 10% a year, double-digit 1 Nov 2022, high-single-digit Aug 2023, inflation +3–4 points (Safran)	\$28.1, Exhibit 4.7, worked example 4.3
R-009	V2500 shop visits 2025: "slightly more than 800" (RTX CFO via Aviation Week, 2026) against about 1,400 forecast (Aviation Week data tool, 2024/25)	\$28.6
R-014	Cost shares: material 60–70% / labour 20–30% / repairs 10–20% (Air Cargo Week) against LLPs 40–60% of parts / high-pressure turbine (HPT) blades and vanes 15–25% / labour 15–20% (Safe Fly)	\$29.1, Exhibit 4.10, worked example 4.4
R-015	HPT blade set: \$1.6M or \$3.08M (per-blade price × 80, unverified) against \$0.18–0.63M implied by the 15–25% share on the two visit ranges	\$29.3
R-022	Used-material price: 60–80% of new on average (Aviation Week) against 70–75% of list from a trader (Aircraft Commerce)	\$28.4, Exhibit 4.10, worked example 4.4
R-013	Shop-visit cost: PR \$1.2–1.6M (this Part) and \$1.8–2.5M (Part I); heavy with LLPs \$2.1–2.8M and ">\$3.5M"; full overhaul \$3.5–4.2M; light \$650–900k (all Safe Fly, consultancy blog)	\$27 worked example 4.2, \$29.4, §30, Exhibit 4.11
D2 T4	Full LLP set "north of \$3m" (Leeham, 2017) and "\$4M" (Aircraft Value News) against a "heavy visit with LLPs" of \$2.1–2.8M all-in (Safe Fly), which cannot include a full set at either price	\$30
R-016	Full -7B LLP set: \$4M (Aircraft Value News, Nov 2019; this Part's figure) against \$5.7M (MyAirTrade, June 2025; Part I's figure)	\$30, worked example 4.5

Id	The two (or more) figures	Where in this Part
R-017	LLP cost per cycle: \$200 (on \$4M) against \$285 (on \$5.7M)	§30, worked example 4.5
R-028	CFM56-7B spare rent: about \$100,000 a month (IBA, Apr 2024) against \$42,000–48,000 (Safe Fly, 2026, may be green-time)	§27.6, §30, worked example 4.5, Exhibit 4.11
R-029	CFM56-7B value: half-life market value \$5.7M / \$6.4M (IBA, Sept 2025) against green-time \$2.8–3.4M and teardown \$0.8–1.2M (Safe Fly, 2026); value of a freshly restored engine unknown	§30, worked example 4.5
R-024	HEICO PMA approvals a year: 300–500 (one reading of the FY2025 10-K) against 400–550 (another)	§29.2, definition box
R-026	FTAI–Chromalloy: "a joint venture" (10-K) against "an exclusive perpetual partnership" (investor commentary); Advanced Engine Repair JV identity unknown	§29.2, definition box

Unknowns carried in this Part

- **U-D2-01 (D2) — U1. Catalog escalation 2024, 2025, 2026.** Only the Nov 2022 (double-digit) and Aug 2023 (high-single-digit) increases and Safran's "inflation + 3–4 points" rule were found. Resolve with Aviation Week/Aircraft Commerce coverage of each year's CFM price revision, or GE/Safran earnings call Q&A on "price".
- **U-D2-02 (D2) — U2. Shop-visit counts 2020–2023.** Only the 2019 level (~2,000+) and the 2024 forecast ("topping 2,000") were found. Resolve from Safran capital-markets-day decks (the safran-group.com media downloads 397180 / 448184 returned in search but not read) or the GE Aerospace Bernstein presentation of 27 May 2026.
- **U-D2-03 (D2) — U3. OEM versus independent share of CFM56 shop visits.** No estimate found. Resolve with Aviation Week/Cirium shop-visit-by-provider data or the Oliver Wyman fleet forecast's provider split.
- **U-D2-04 (D2) — U4. CFM56 share of GE CES services revenue.** Not disclosed in the 10-K. GE has historically described the CFM56 as the largest services contributor; a number would need an investor presentation or call transcript (the 22 Jan 2026 webcast transcript URL was returned, not read).
- **U-D2-05 (D2) — U5. Capacity of the named airline MROs.** AFI KLM E&M, Delta TechOps, Turkish Technic, GMF AeroAsia, EgyptAir M&E, GA Telesis, Lufthansa Technik and ST Engineering: no CFM56 inductions- per-year or expansion figures returned. Resolve via each company's annual report or trade-press shop profiles.

- **U-D2-06 (D2) — U6. Engine-stand shortages and HPT blade / casting / forging lead times.** No numbers found. Resolve via Aviation Week supply-chain coverage, PCC/Howmet commentary, or GE call Q&A.
- **U-D2-07 (D2) — U7. Module-exchange pricing.** No source gives the exchange fee or a worked price for a CFM56 HPT or core exchange; Safe Fly's "\$150,000–\$400,000 removal cost per module" is ambiguous. D5 may find it in FTAI's investor materials or the In Practise interview ("FTAI Aviation: CFM56 repair and module swap", URL returned, not read). Cluster K-01 with U-D3-20 ("U20 — Pricing of a module exchange (exchange fee, LLP-life true-up) from any provider, including FTAI.") and U-D5-03 ("U3 — Exchange turnaround time and exchange pricing.").
- **U-D2-08 (D2) — U8. Value of a freshly overhauled full-stack CFM56-7B** versus a green-time unit. Needed to close the worked example. Resolve via IBA/Ascend/mba engine value tables. Cluster K-02.
- **U-D2-09 (D2) — U9. IAE "FleetCare".** Not found as a current product; only EngineWise. Resolve on the P&W services site.
- **U-D2-10 (D2) — U10. Dates.** Several Aviation Week articles were returned without publication dates (noted "c. 2023/24" or "2025"). The writer should confirm before quoting.
- **U-D1-10 (D1) — U10** Years and amounts of each CFM LLP escalation step 2019–2025 (only "Aug 2023, +20–30% on -7B parts" and "~12% LLP escalation" without a year were reached). Cluster K-03 with U-D2-01.
- **U-D1-07 (D1) — U7** Labour hours per workscope and shop labour rates (US\$/hour) for CFM56 visits. Resolve with an MRO price catalogue or Aircraft Commerce's maintenance budgets.
- **U-D1-14 (D1) — U14** Whether a module exchange requires a test-cell run under the CFM56 ESM.
- **U-D3-15 (D3) — U15** — GE Aerospace's used-material arm: name, scale and CFM56 role not captured (the only hit was a TrueEngine presentation URL whose content was not read). Cluster K-16 with U-D4-10.
- **U-D4-10 (D4) — U10 — Any CFM/GE statement specifically on the Chromalloy HPT blade PMA,** and any GE "material" programme (e.g. TrueChoice Material) positioning against PMA. The brief's phrase "material solutions" was not matched to a current CFM/GE programme in this session; the historical P&W "Global Material Solutions" CFM56-3 PMA venture was found only via headlines.
- **U-D4-14 (D4) — U14 — Terms of the FTAI–CFM Jan 2026 materials agreement** (duration, volumes, pricing, whether it restricts FTAI's PMA use). Cluster K-19 with U-D5-18 ("U18 — Terms of the CFM materials agreement mentioned on the Q4 2025 call.").

- **U-D4-16 (D4) — U16 — CFM56-7B HPT blade count per set** (80 is a recollection, unverified here) and the OEM list price for a full HPT blade and vane set.
- **U-D5-05 (D5) — U5 — Denominator of the 25% share goal** and the year by which it is targeted.
- **U-D5-08 (D5) — U8 — GMF AeroAsia and EgyptAir terms** (capacity committed, economics, duration).
- **U-D6-07 (D6) — U7 — Security deposit and maintenance deposit balances.** FTAI's balance-sheet "security deposits" and "maintenance deposits" liabilities by year, and the reserve rates in its leases. Resolve: FY2025 10-K balance sheet and revenue-recognition note. (Carried here for the green-time reserve rates Option C needs.)
- **U-D9-08 (D9) — U8 — Labour figures (engine MRO technician headcount, shortfall, wage inflation).** Oliver Wyman cites skilled-worker retirements qualitatively. D2 may hold these.
- **U-D11-16 (D11) — U16 — GE CFM56 shop-visit count, spare-parts price increases, and any "material solutions" or used-serviceable-material programme wording.** Not returned from the 10-K. Resolve: read the CES segment discussion in the 10-K and the Q4 2025 and Q2 2026 earnings materials.

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